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Spaulding et al.

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(54) **FAN ASSEMBLY**

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(73) Assignee: **MILWAUKEE ELECTRIC TOOL CORPORATION**, Brookfield, WI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Related U.S. Application Data

(60) Provisional application No. 63/294,935, filed on Dec. 30, 2021, provisional application No. 63/160,305, filed on Mar. 12, 2021.

(51) **Int. Cl.**
F04D 29/52 (2006.01)
F04D 25/06 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **F04D 29/522** (2013.01); **F04D 25/0673** (2013.01); **F04D 25/08** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC ... A47B 13/081; F16D 41/185; F04D 19/002;
F04D 25/0673; F04D 25/084;
(Continued)

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Primary Examiner — Essama Omgba

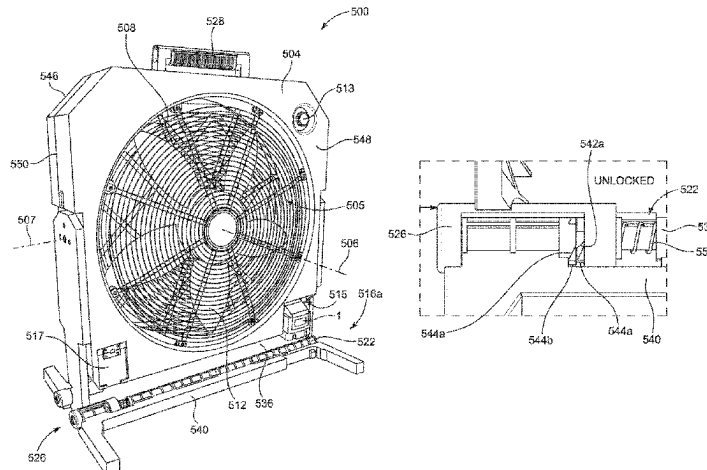
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(57) **ABSTRACT**

A battery powered box-type fan assembly includes a fan housing that at least partially defines an airflow chamber, a plurality of fan blades positioned within the airflow chamber, a motor operable to rotate the plurality of fan blades, a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor, and a base coupled to the fan housing and configured to be placed on a support surface. The base elevates the fan housing above the support surface and is operable to support the fan housing in multiple configurations. The base also includes a stand pivotable relative to the fan housing between a collapsed position, in which the stand

(Continued)



is flipped up toward the fan housing, and a deployed position, in which the stand extends away from the fan housing and is configured to contact the support surface.

19 Claims, 32 Drawing Sheets

(51) Int. Cl.

F04D 25/08 (2006.01)
F04D 29/60 (2006.01)
F04D 29/64 (2006.01)
F04D 29/70 (2006.01)
F04D 19/00 (2006.01)
F16D 41/18 (2006.01)

(52) U.S. Cl.

CPC **F04D 29/601** (2013.01); **F04D 29/646** (2013.01); **F04D 29/703** (2013.01); **F04D 19/002** (2013.01); **F04D 25/084** (2013.01); **F16D 41/185** (2013.01)

(58) Field of Classification Search

CPC **F04D 29/522**; **F04D 29/601**; **F04D 29/646**; **F04D 29/703**; **F04D 25/08**; **Y10T 74/214**; **B25B 13/46**; **B25B 13/466**
 See application file for complete search history.

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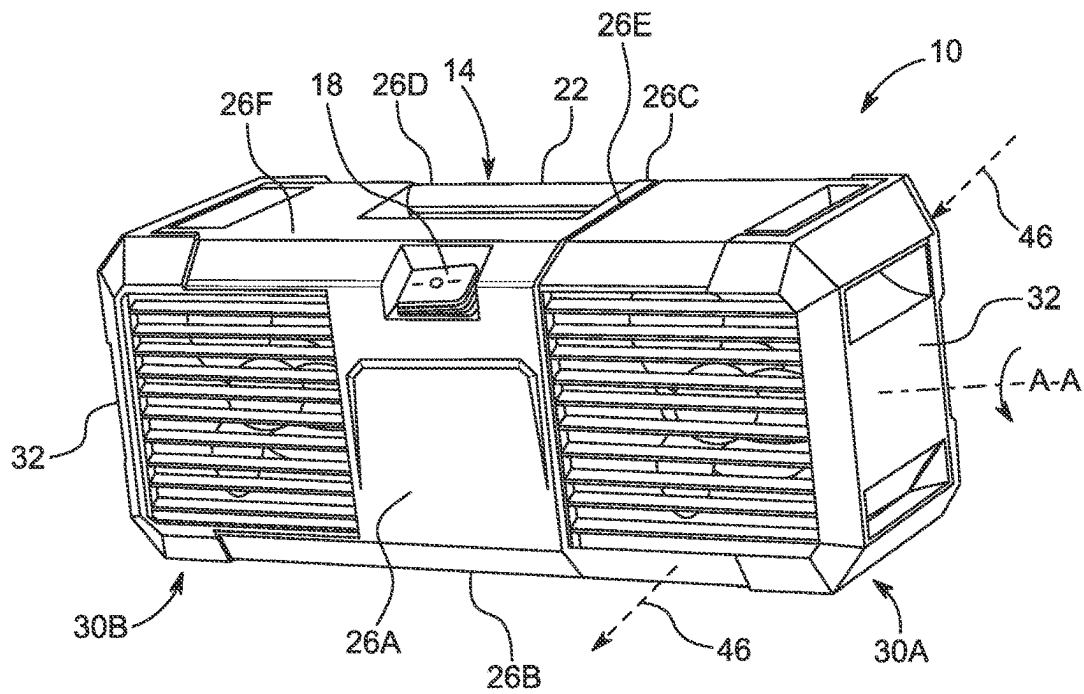


FIG. 1

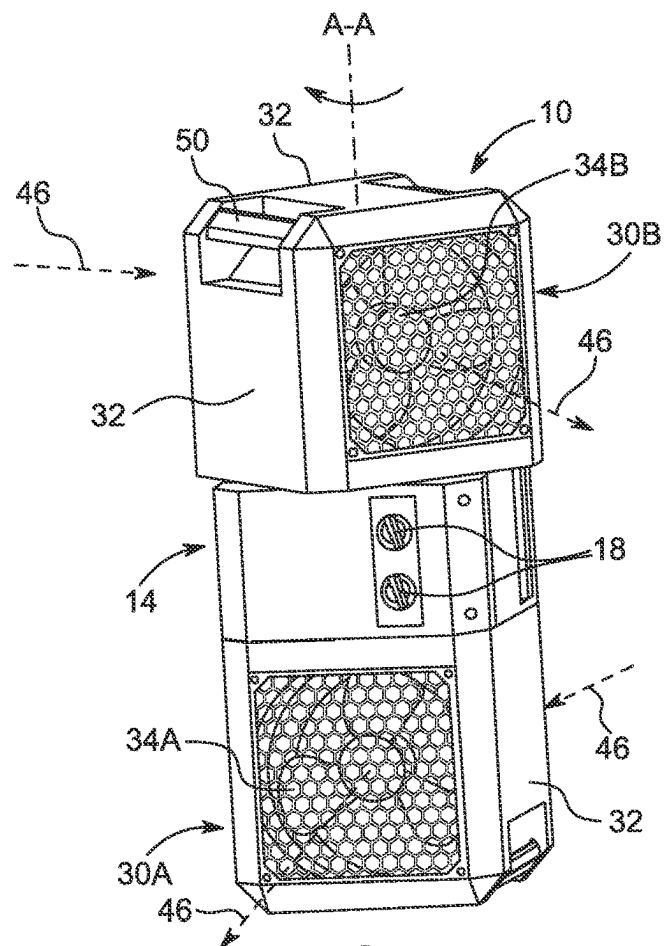


FIG. 2

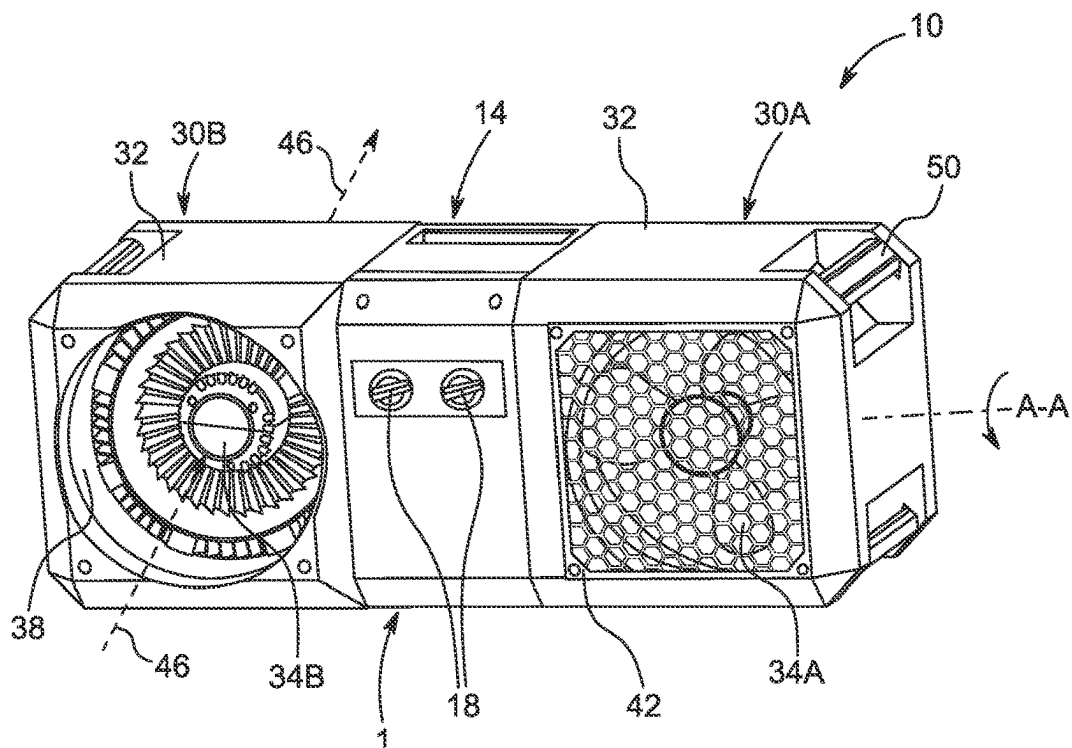


FIG. 3

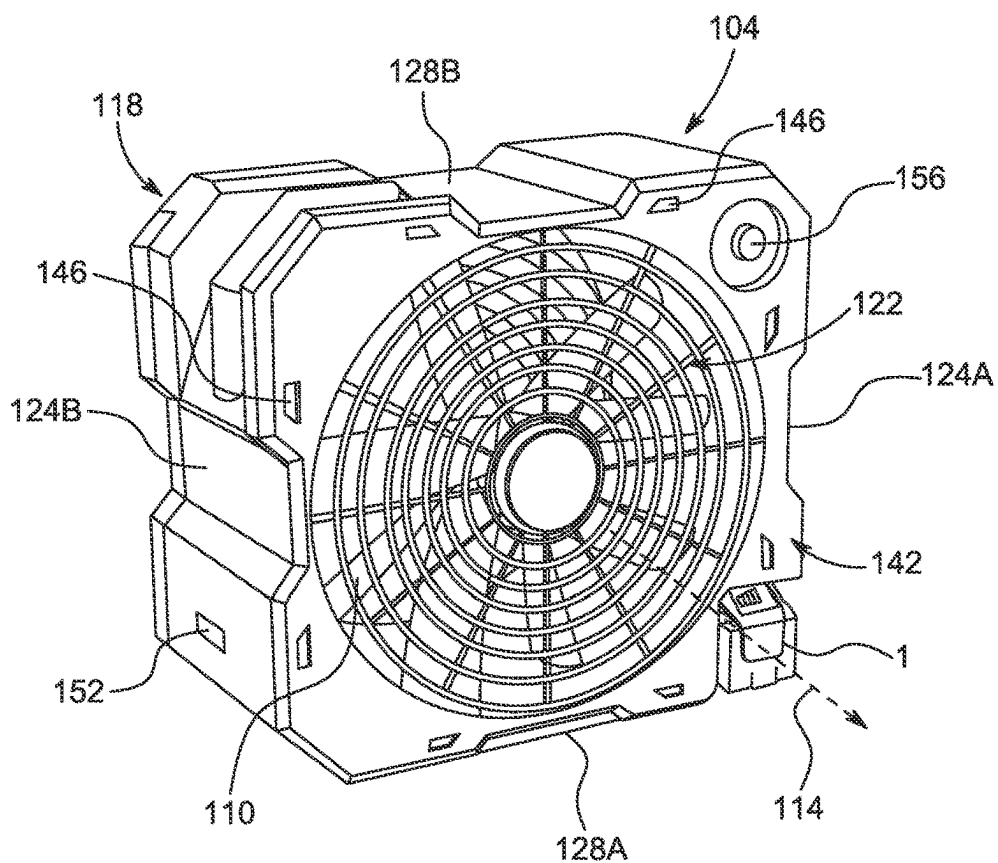


FIG. 4

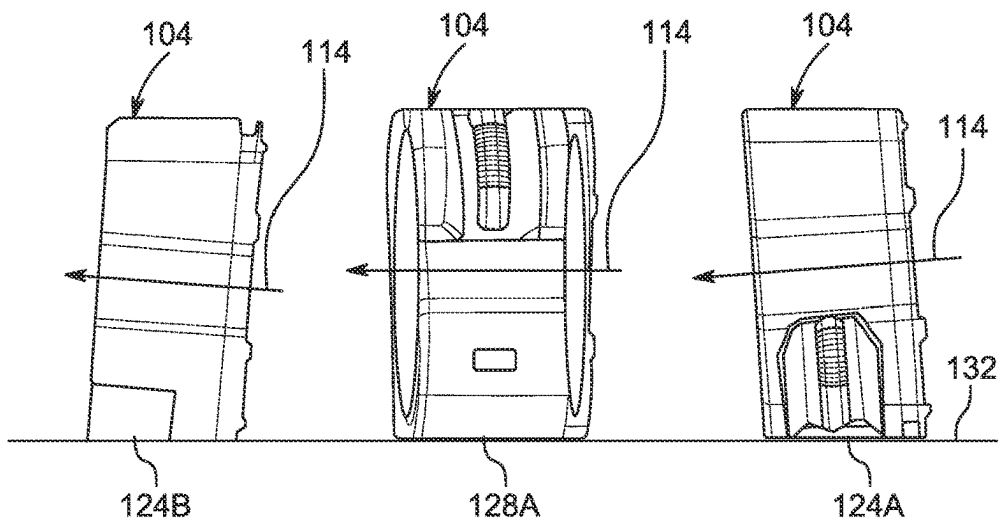


FIG. 5

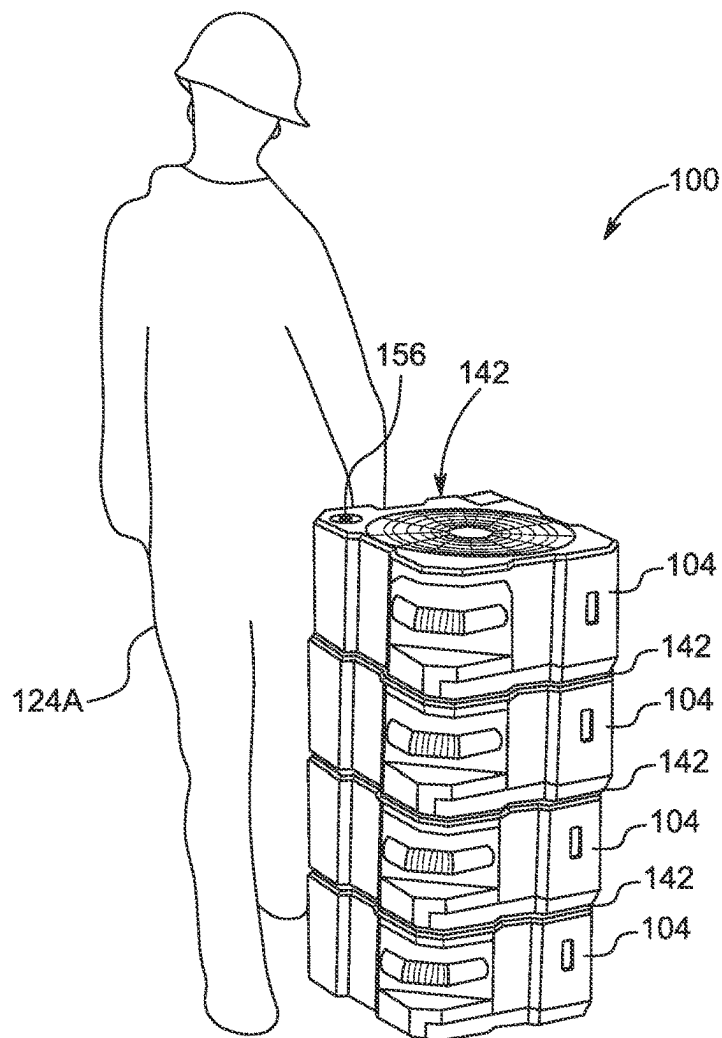


FIG. 6

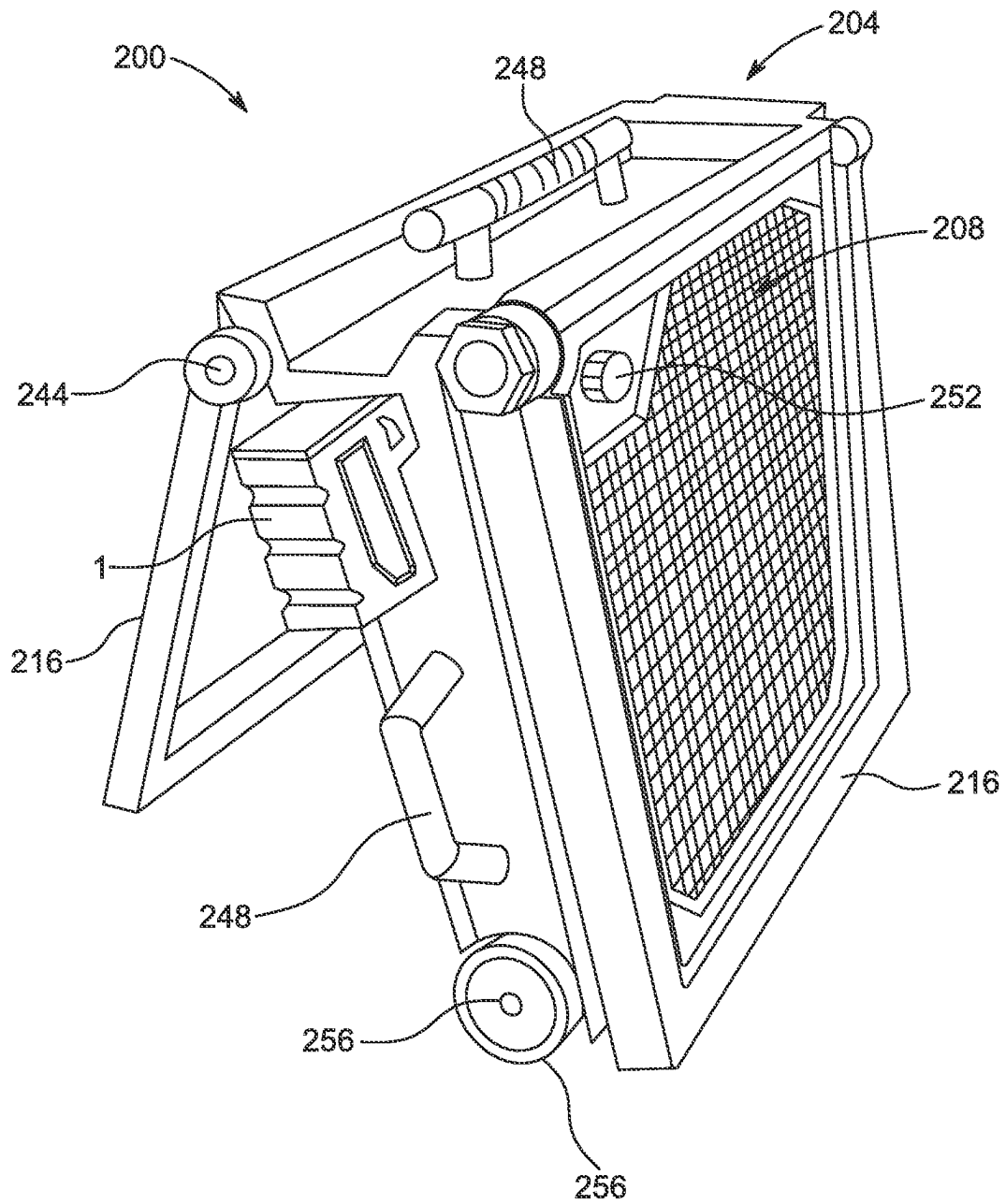


FIG. 7

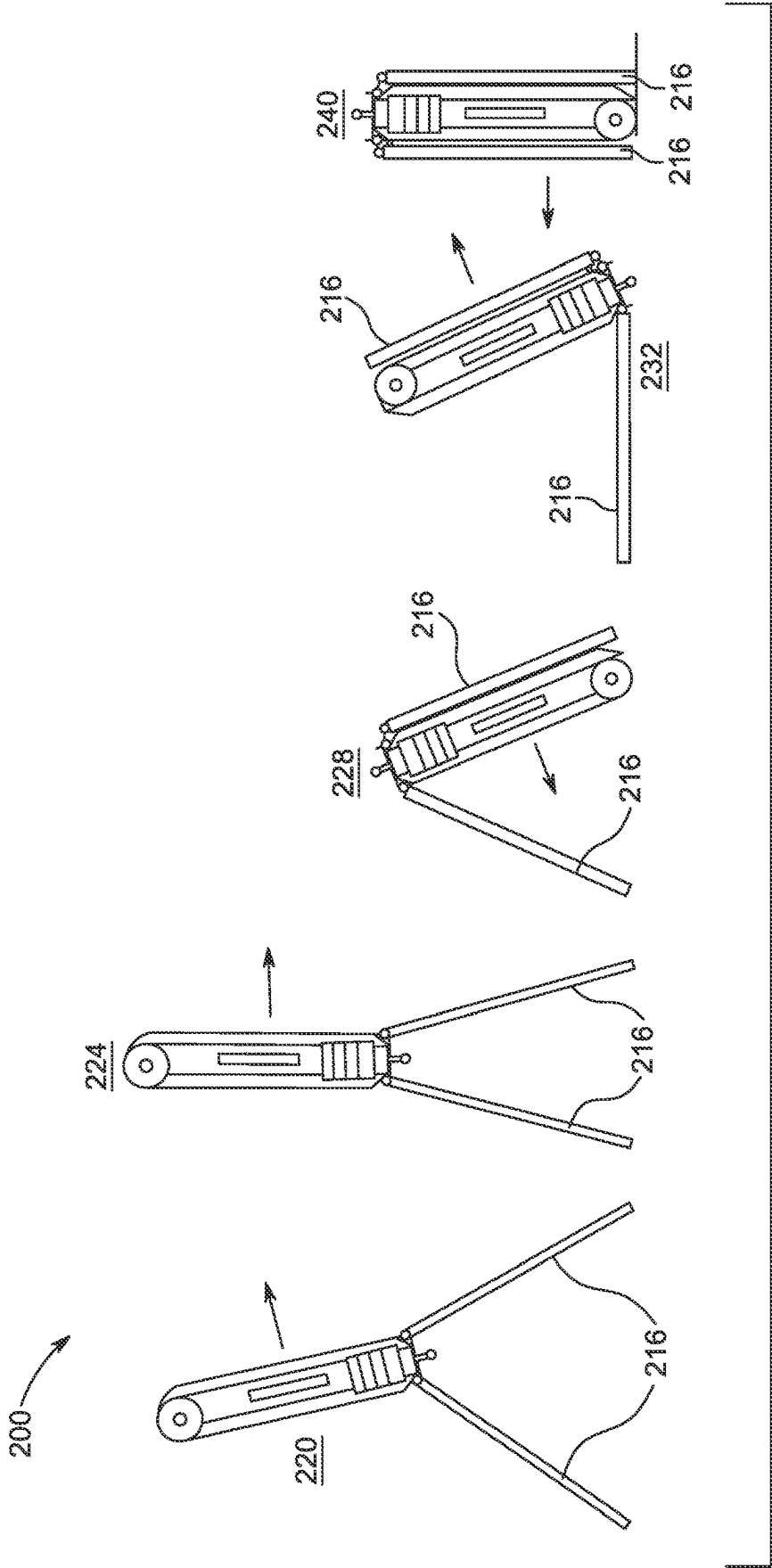


FIG. 8

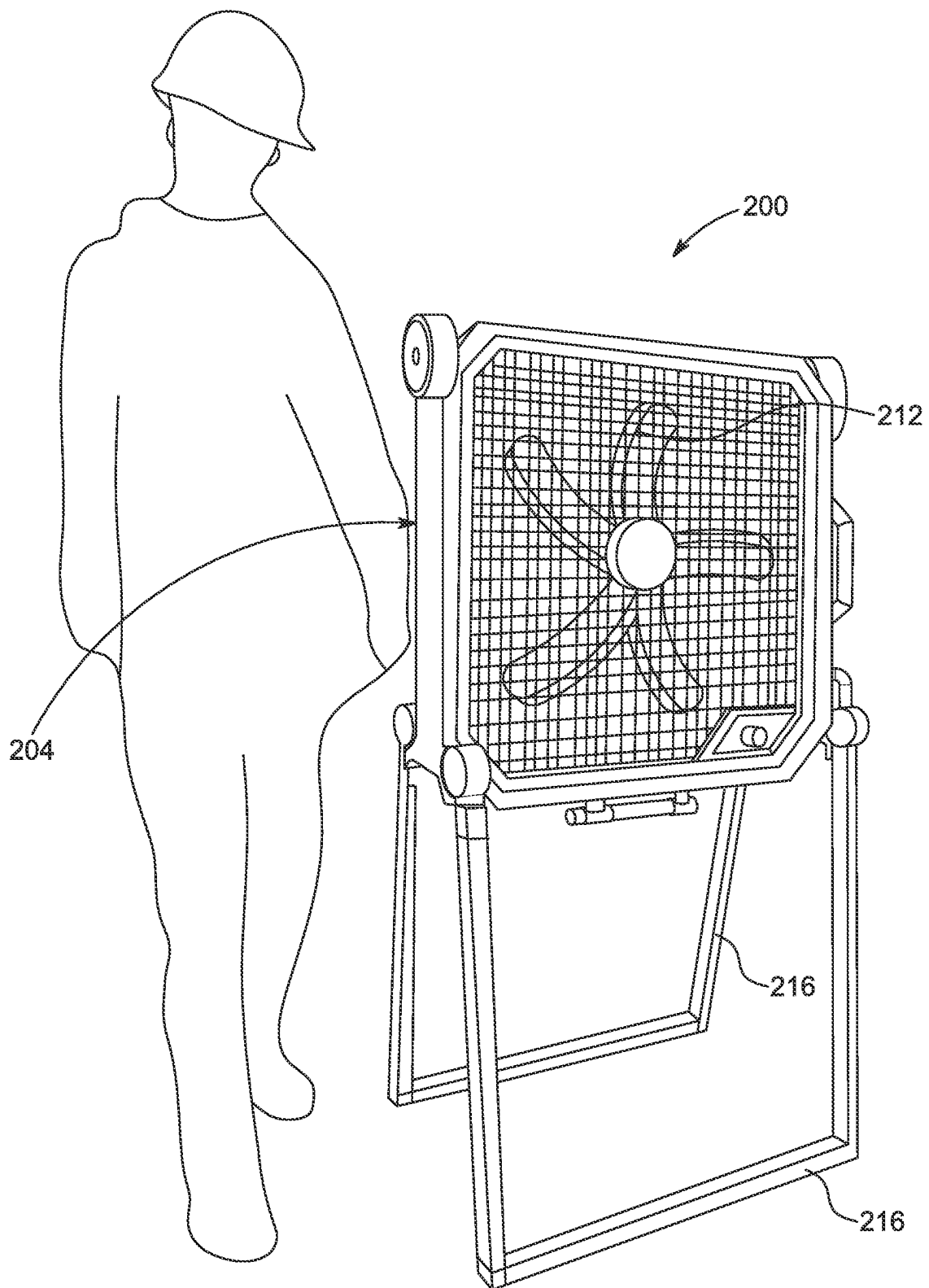


FIG. 9

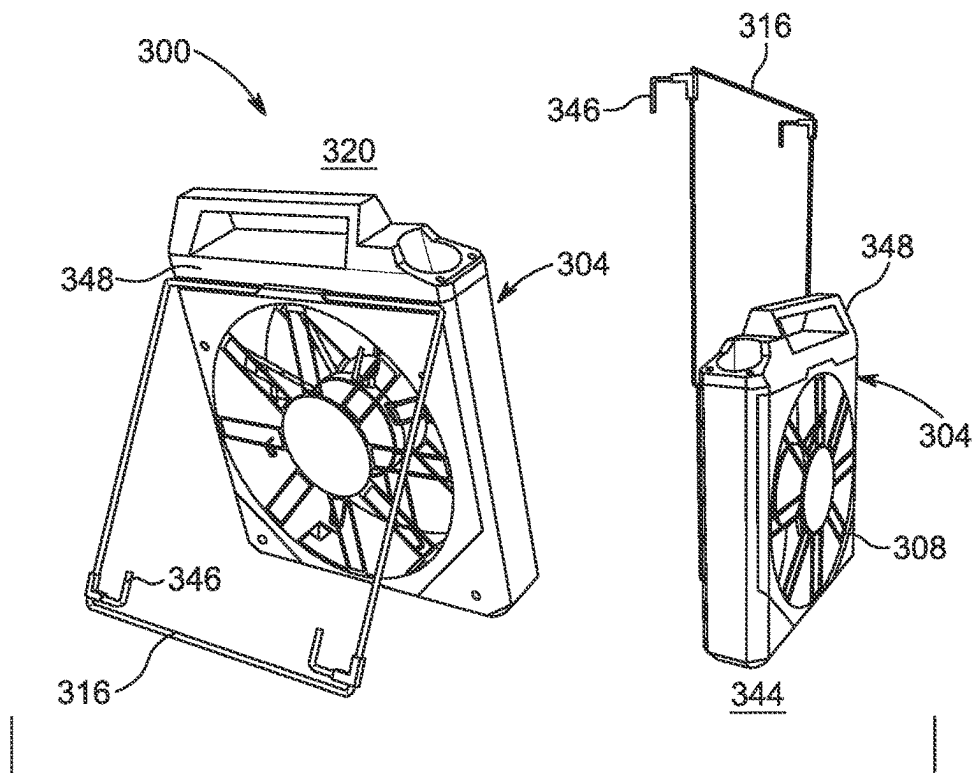


FIG. 10

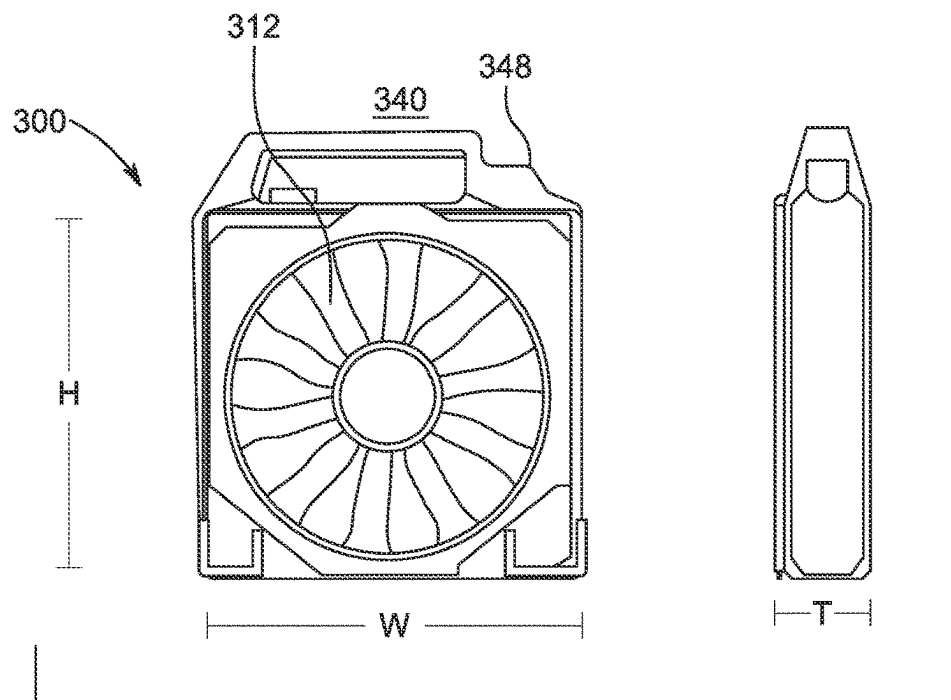


FIG. 11

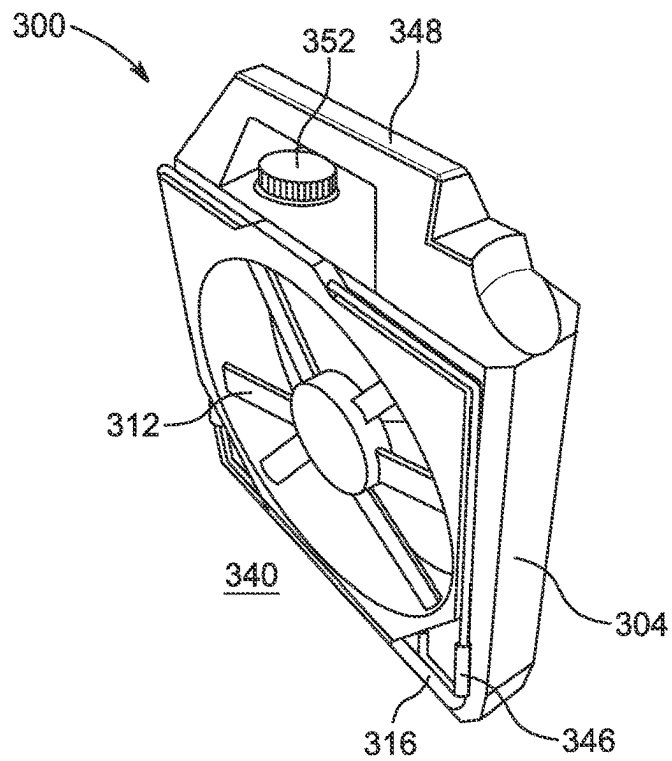


FIG. 12

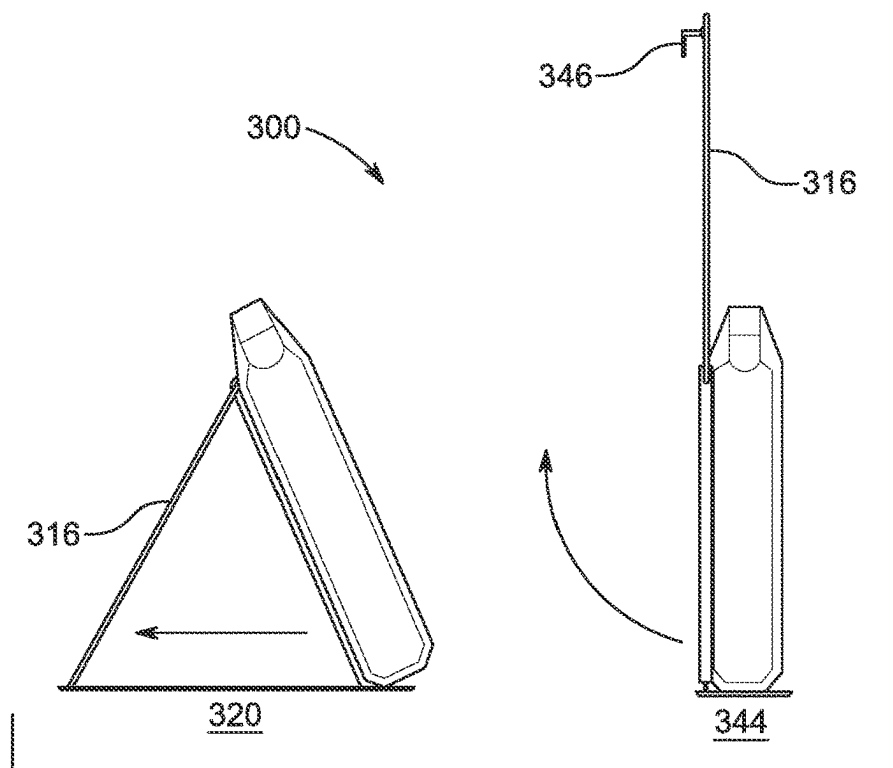


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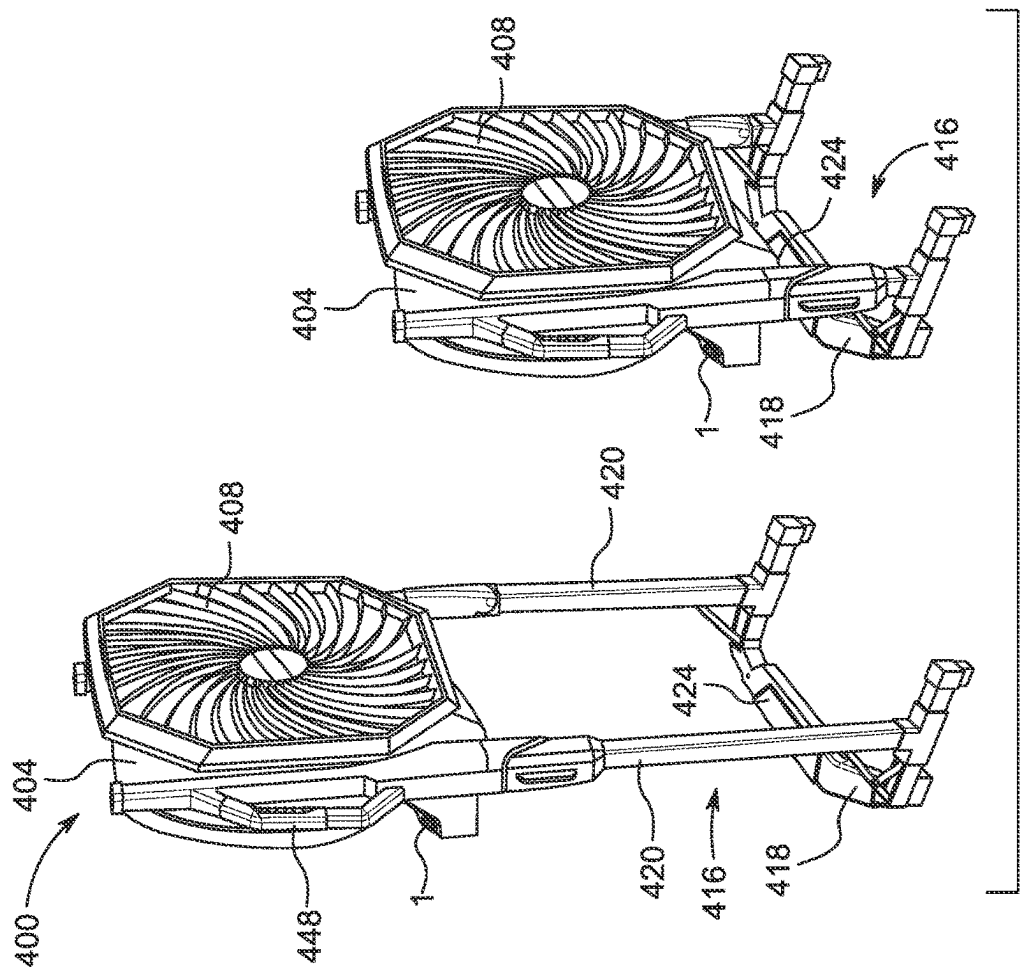


FIG. 14

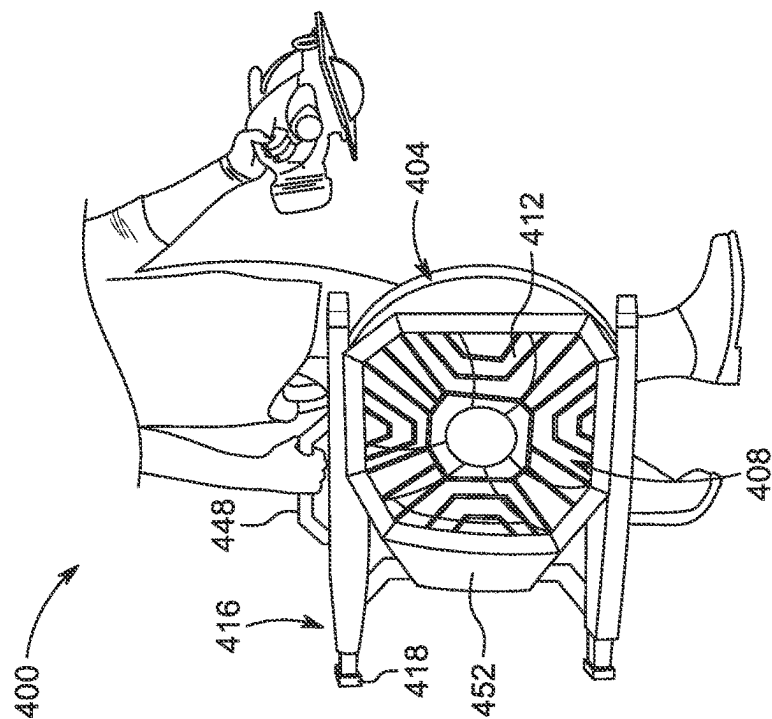


FIG. 15

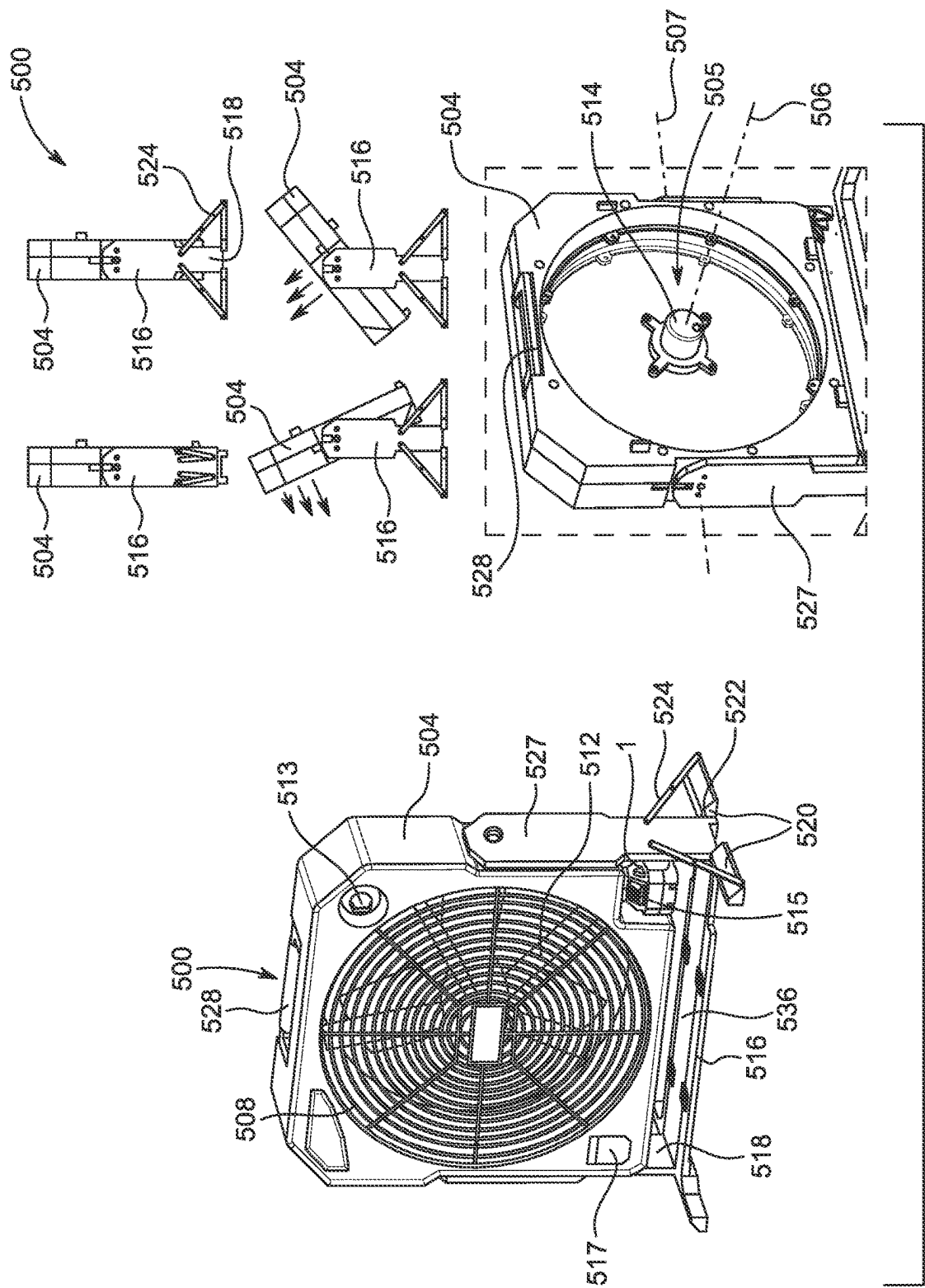


FIG. 16

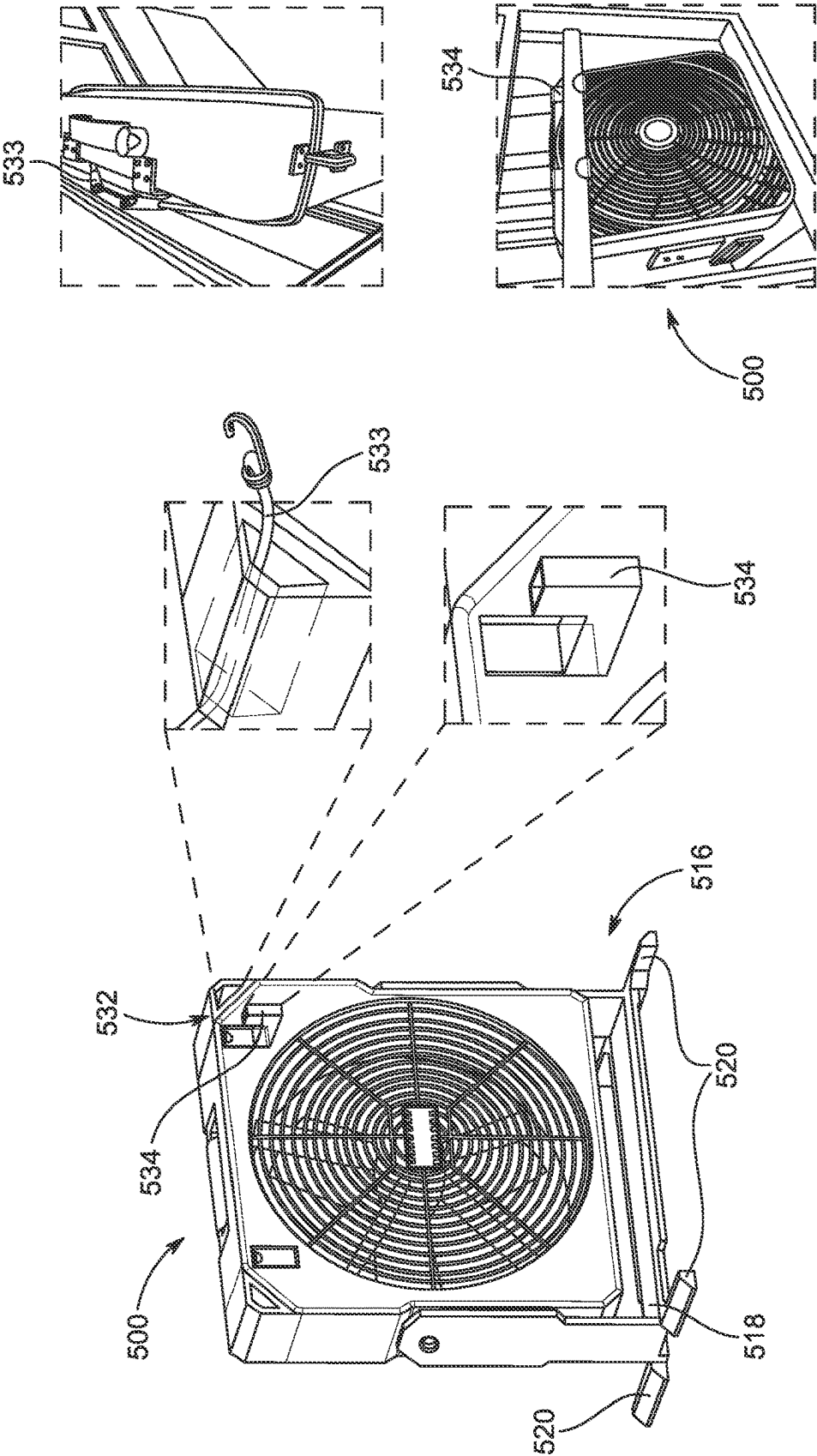


FIG. 17

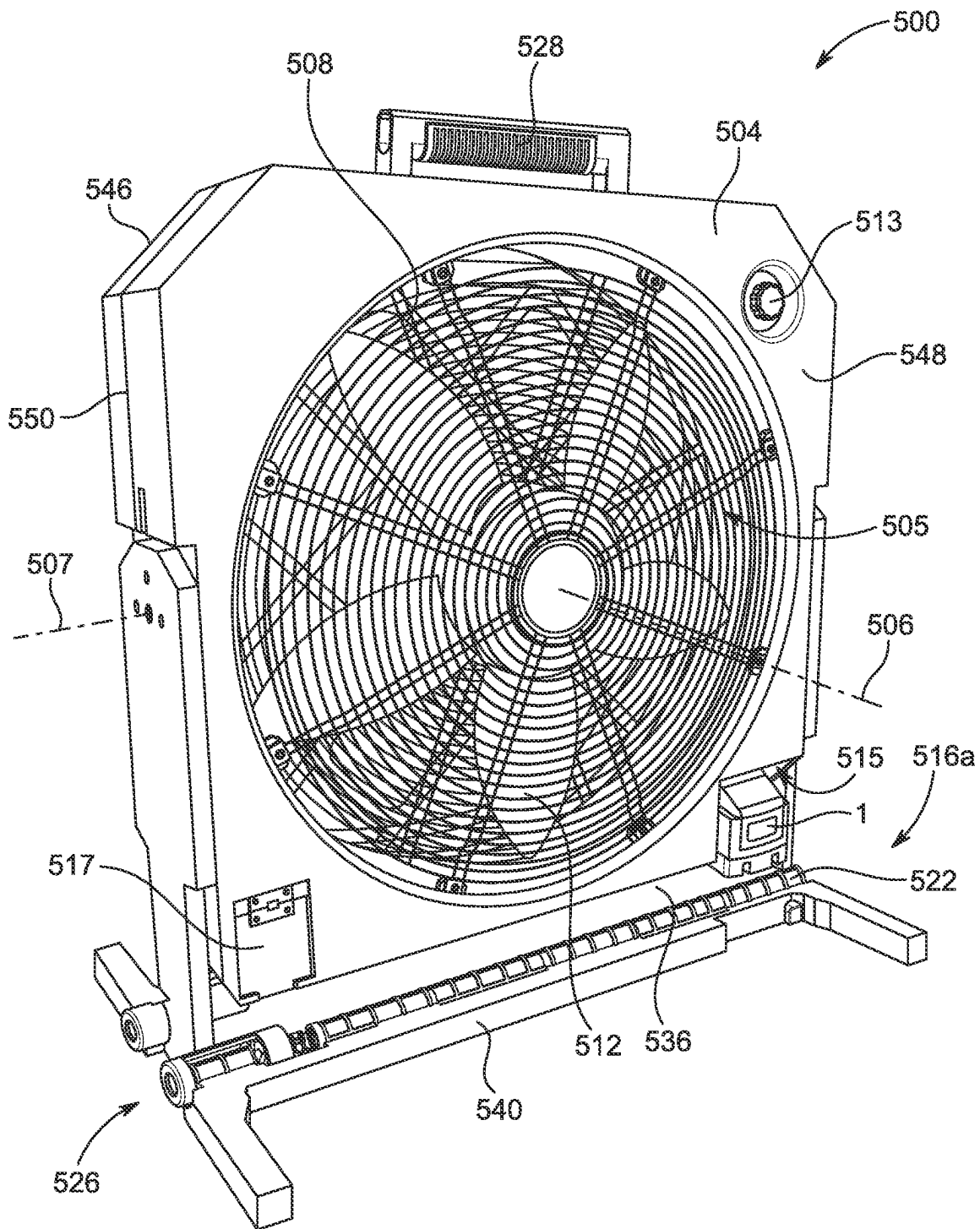


FIG. 18A

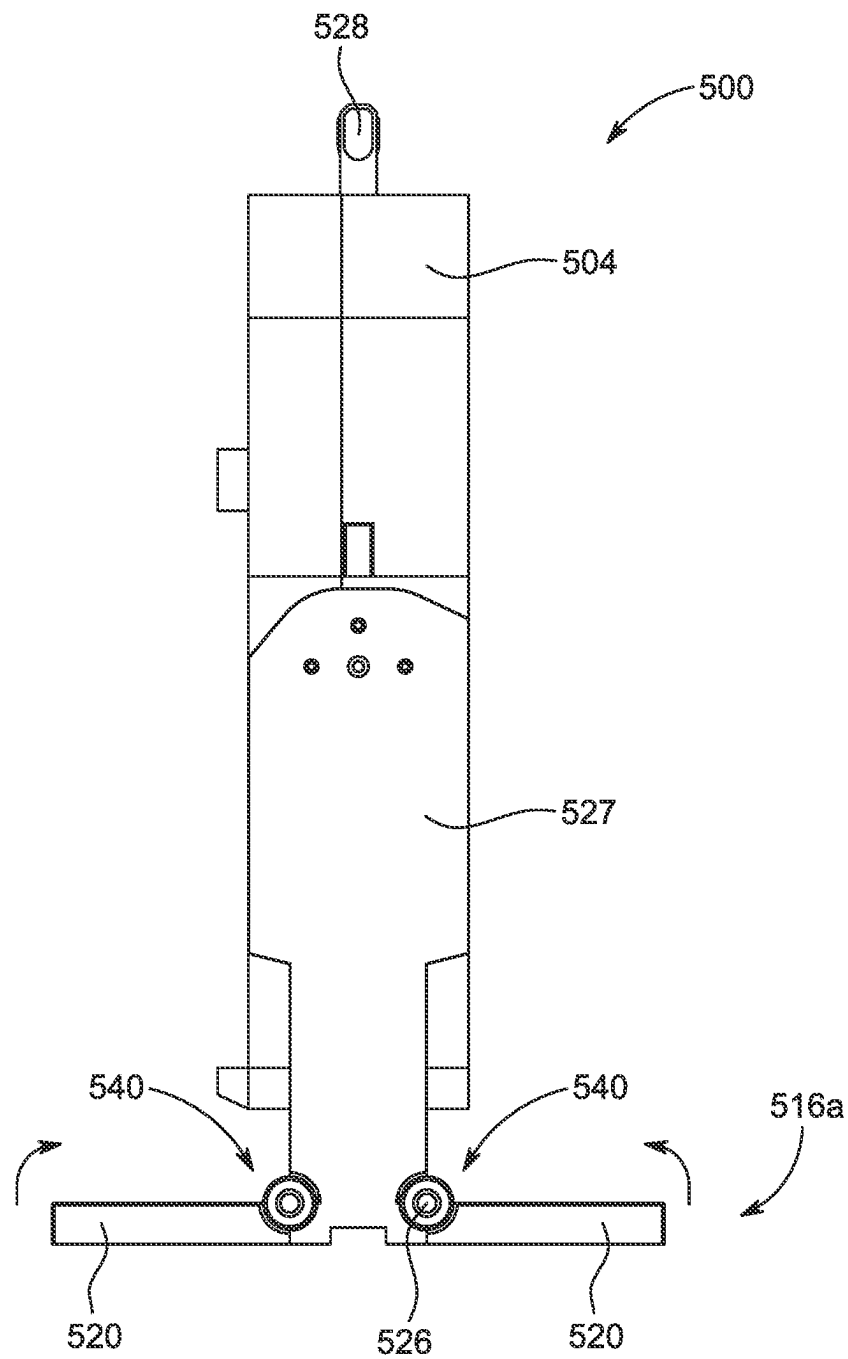


FIG. 18B

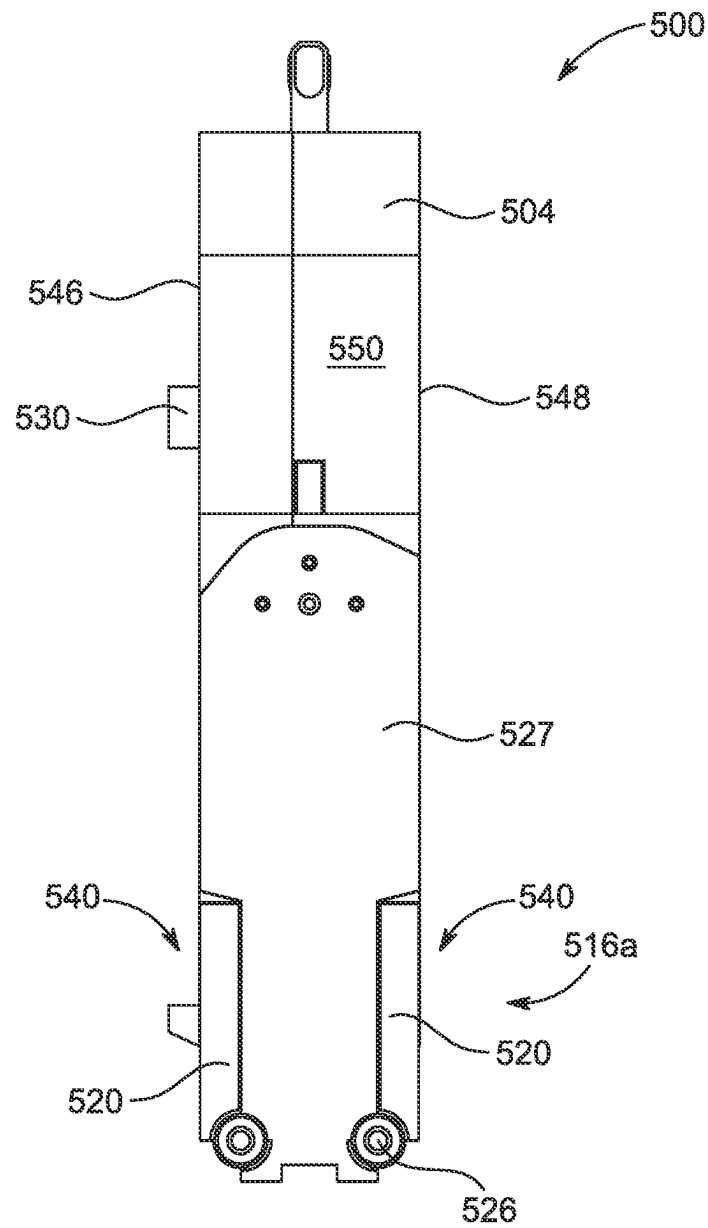


FIG. 18C

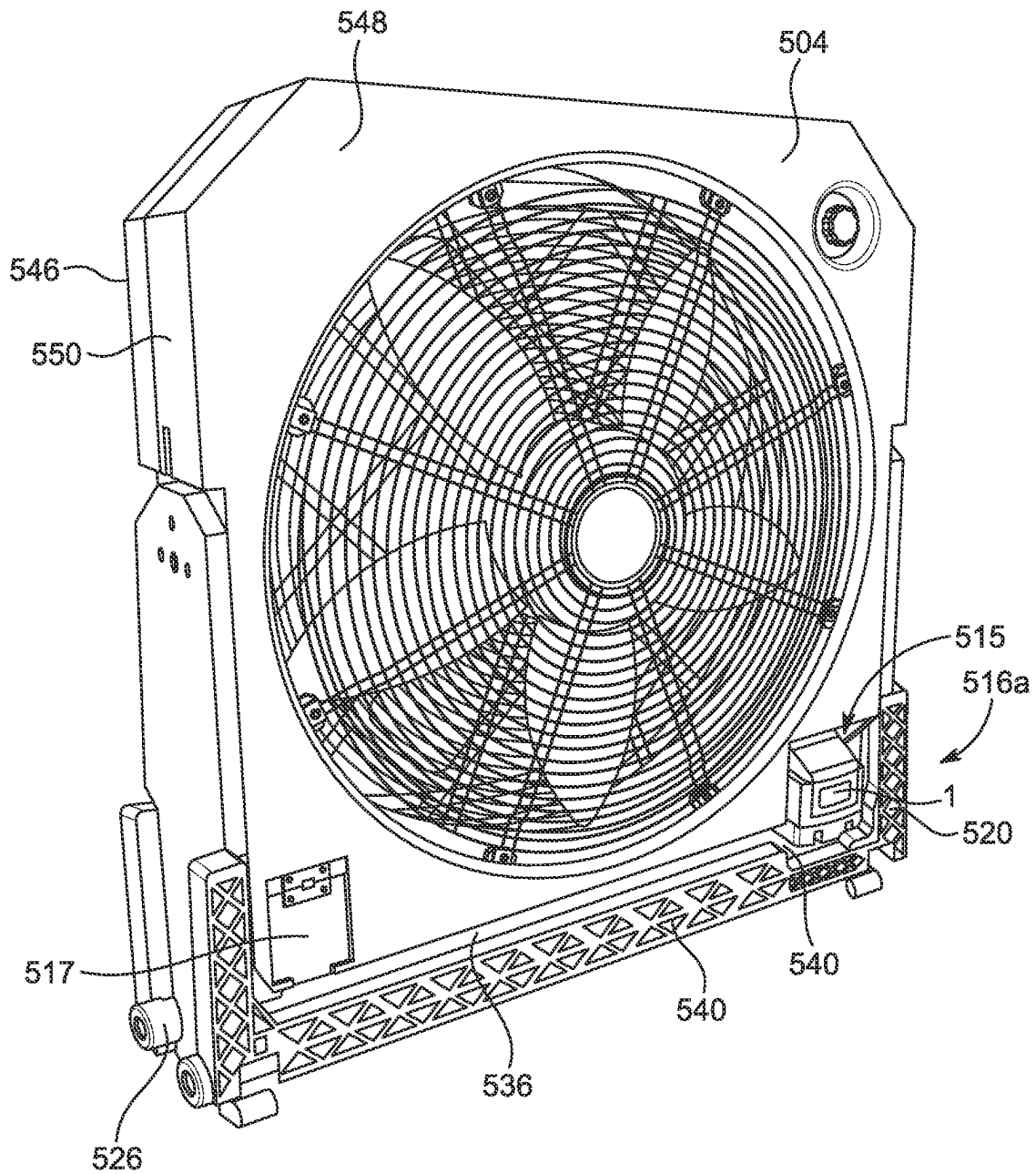


FIG. 18D

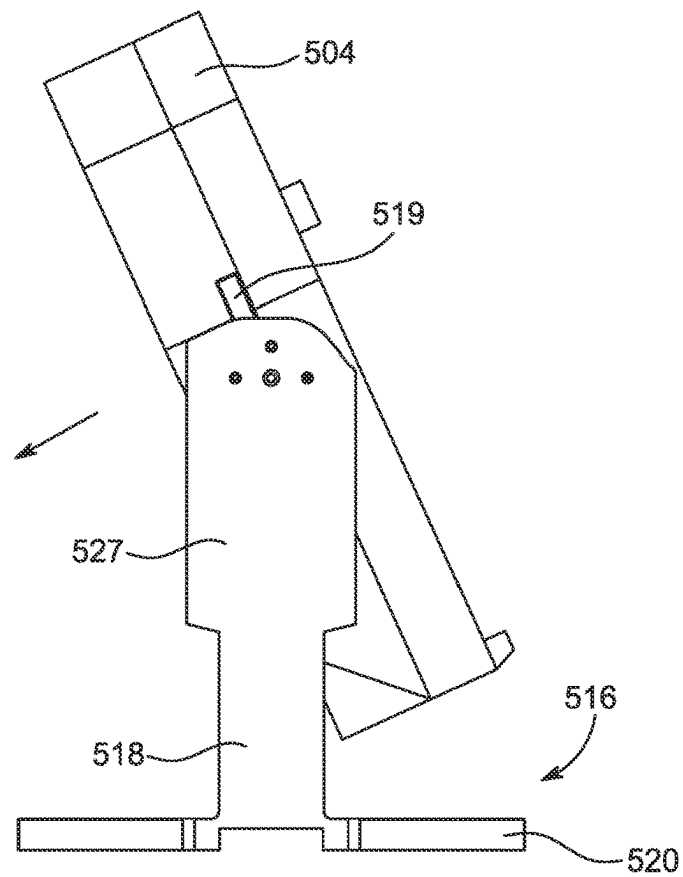


FIG. 18E

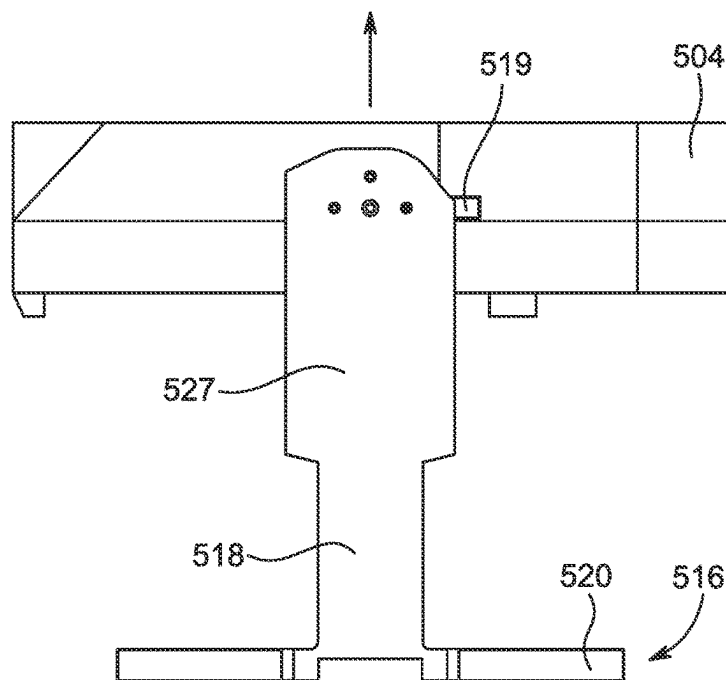


FIG. 18F

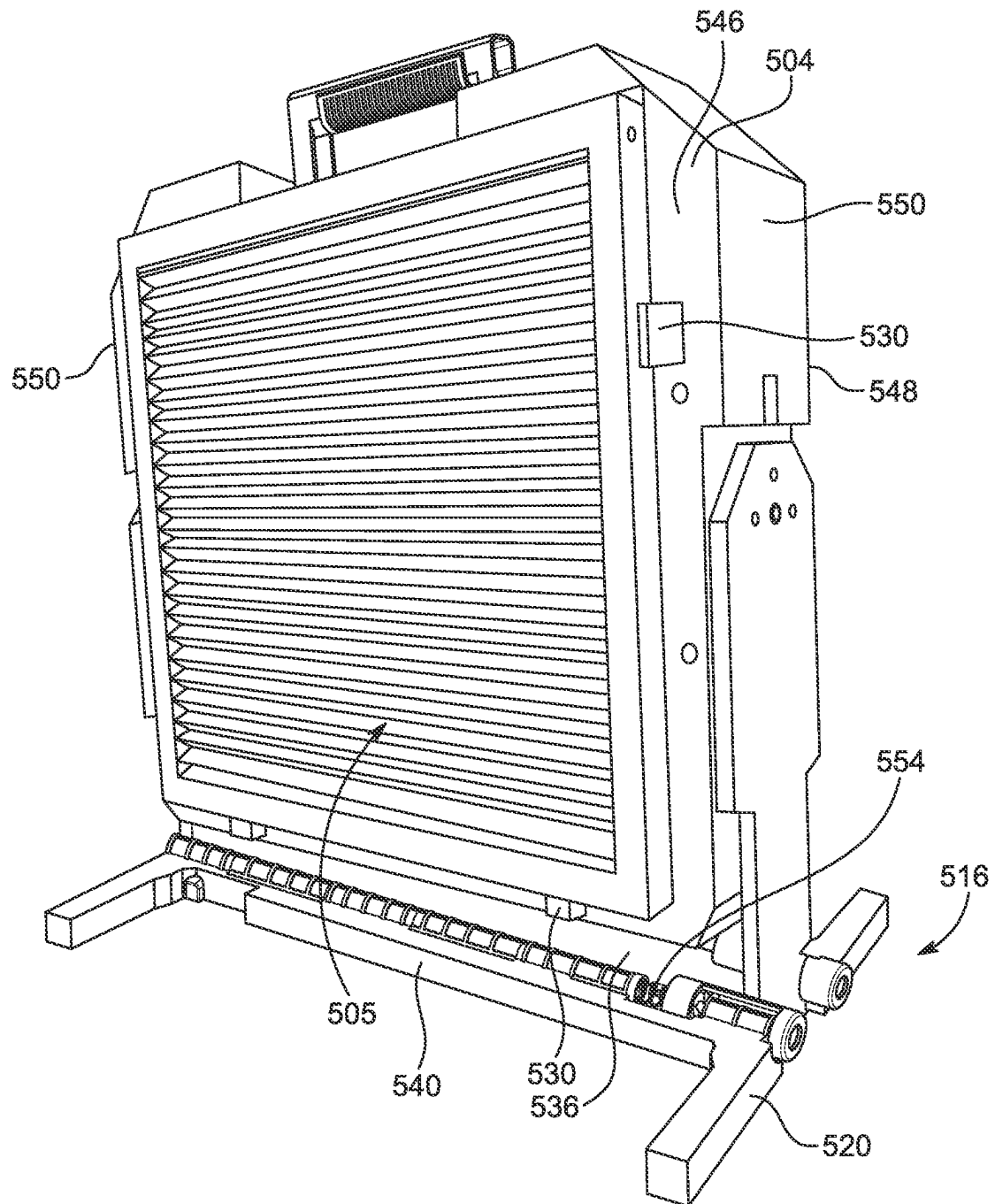


FIG. 18G

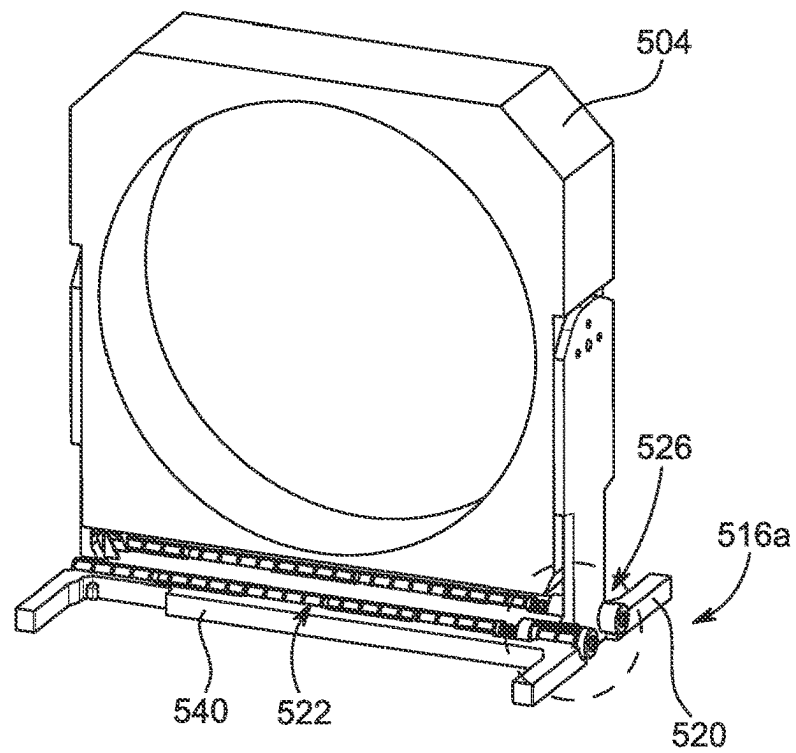


FIG. 19A

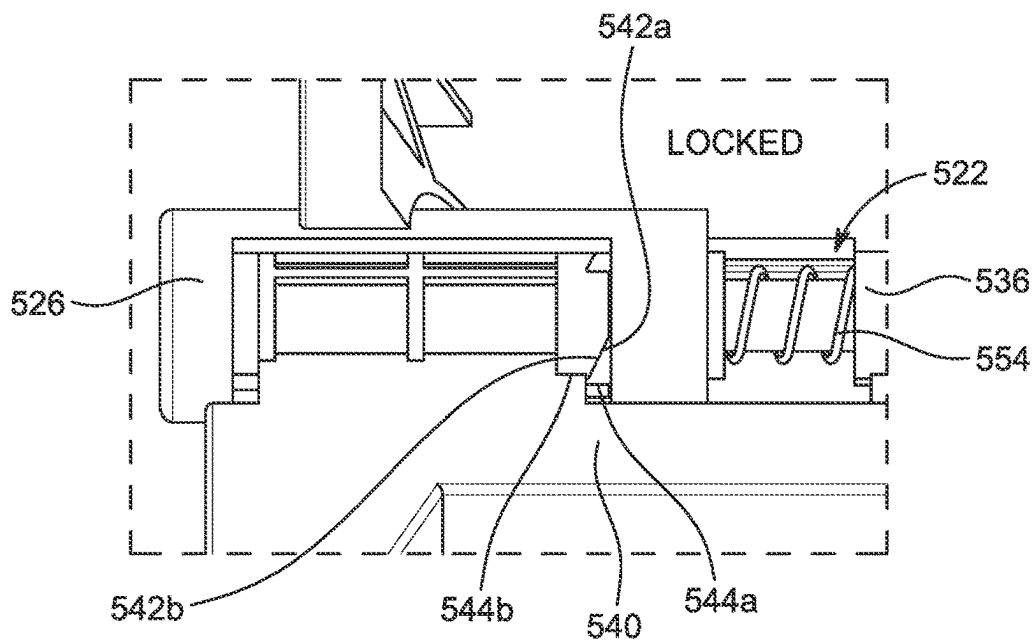


FIG. 19B

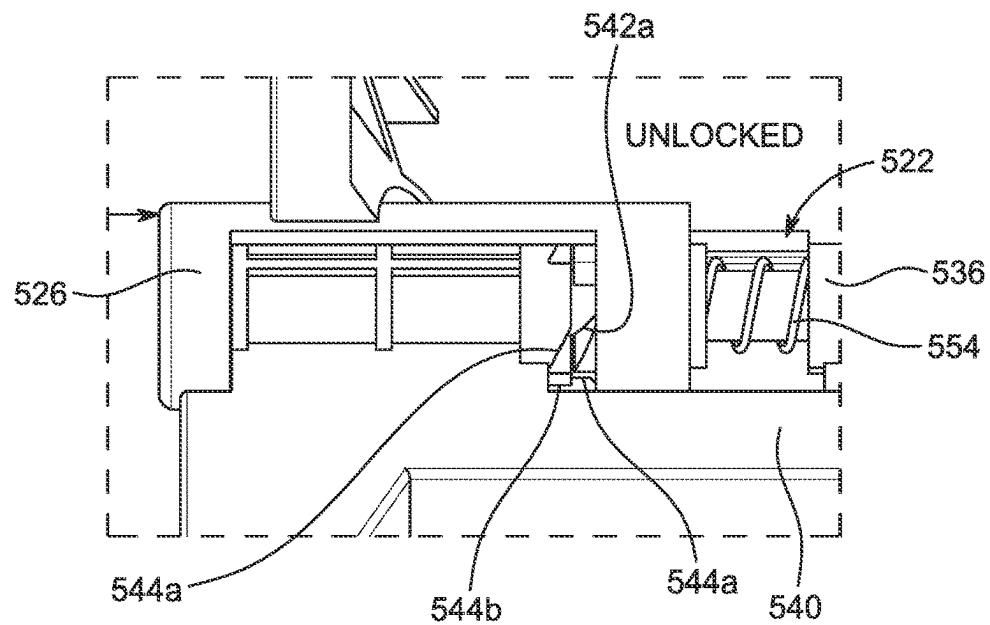


FIG. 19C

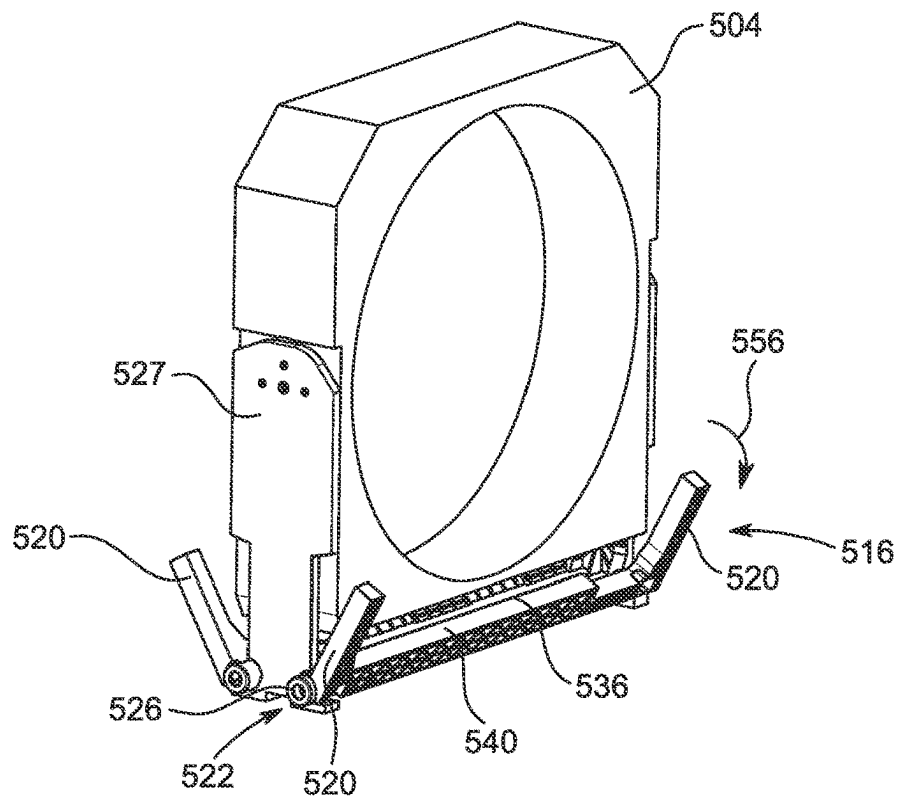


FIG. 19D

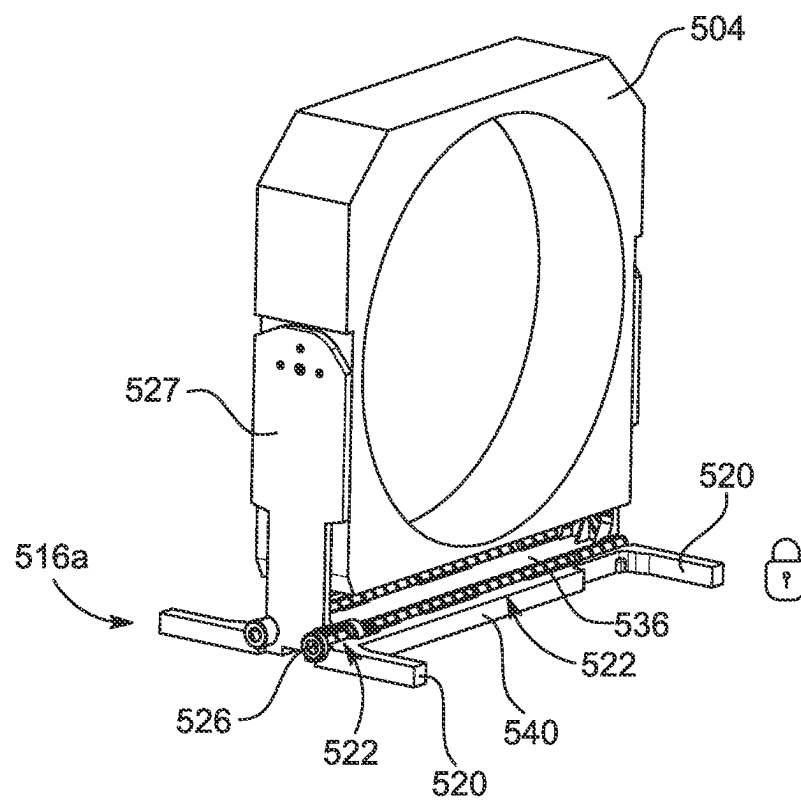


FIG. 19E

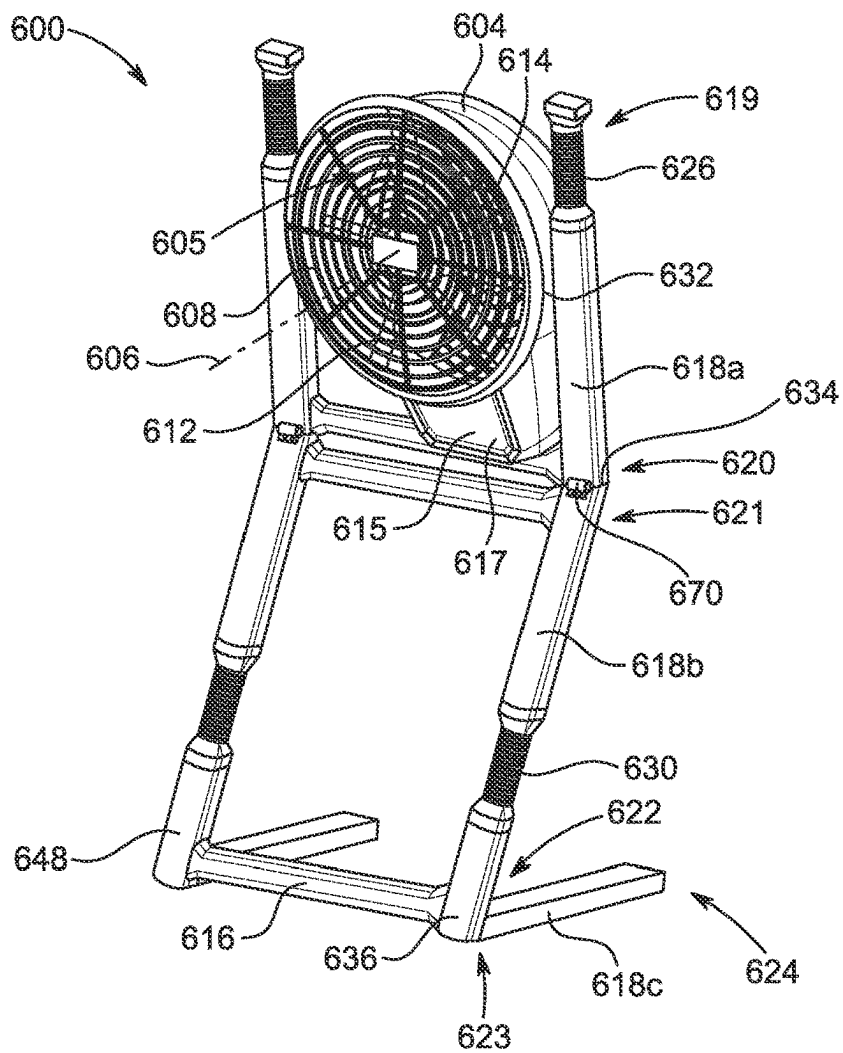


FIG. 20A

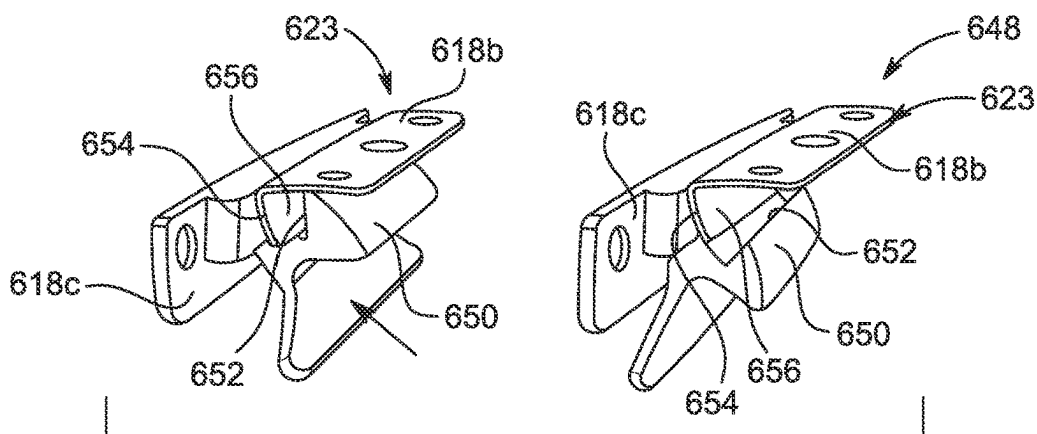


FIG. 20B

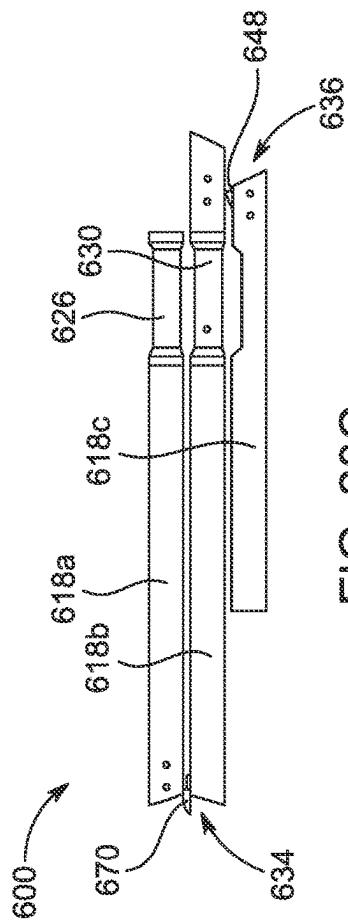


FIG. 20C

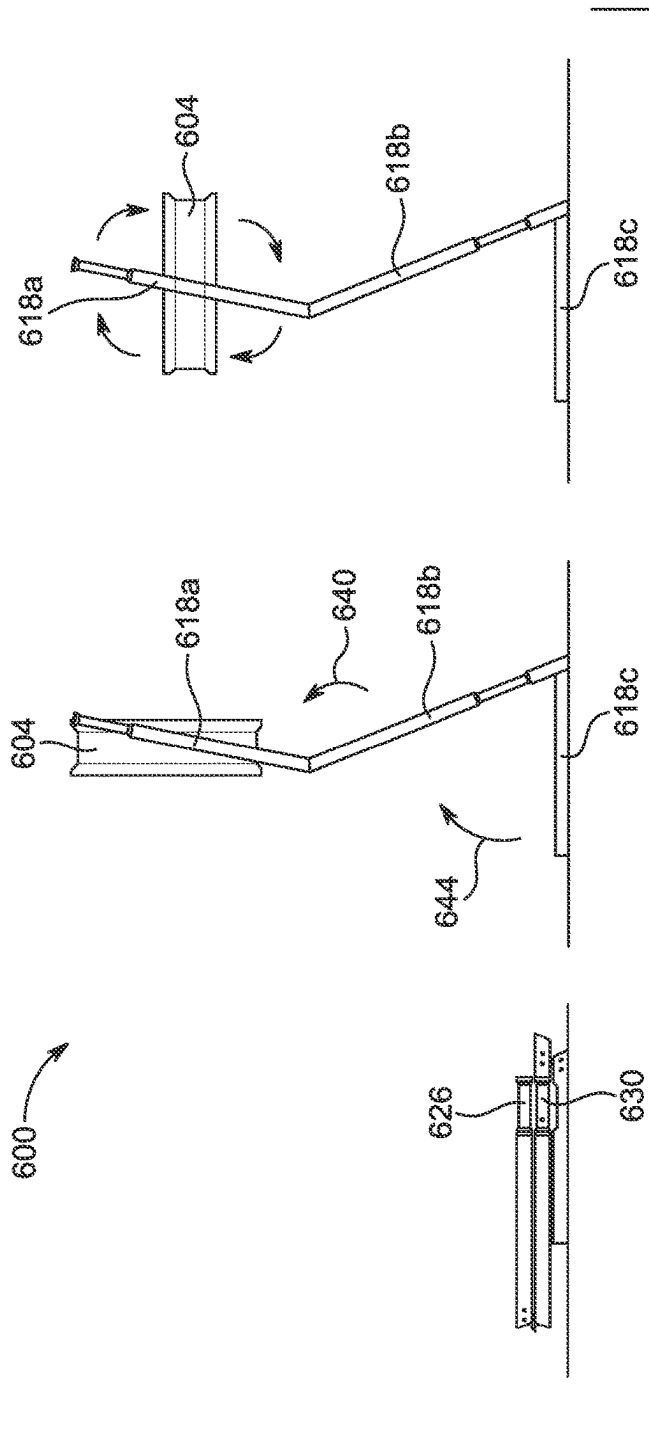


FIG. 20D

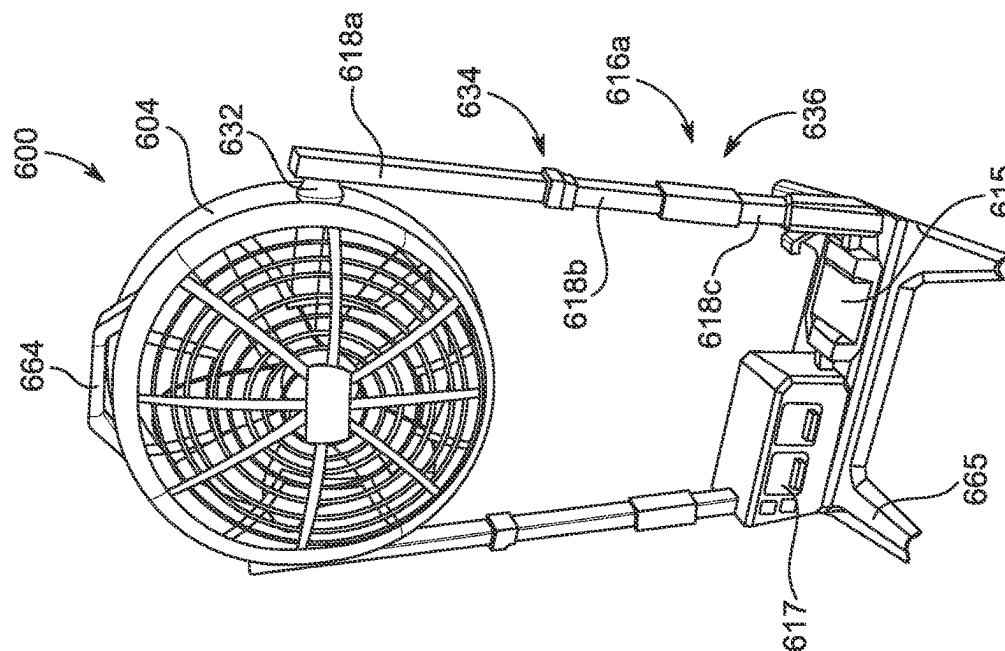


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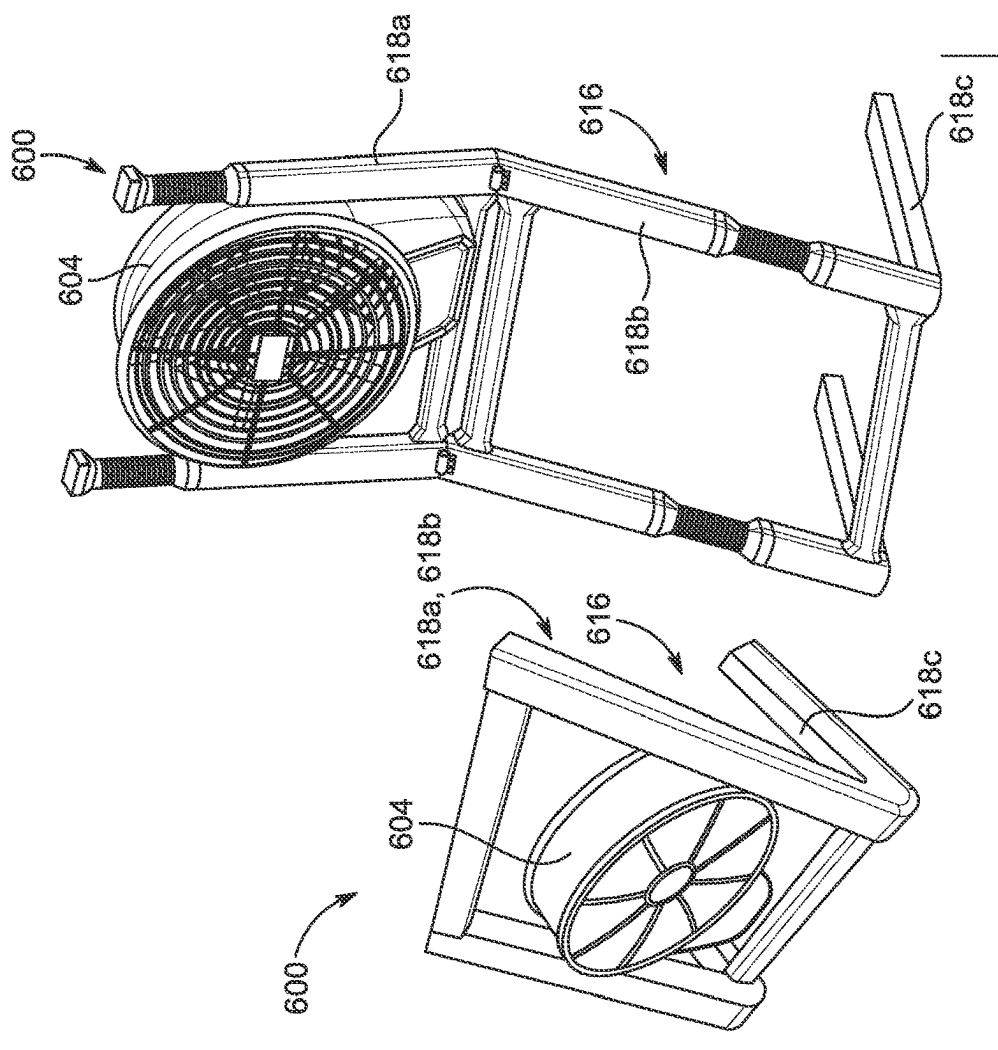


FIG. 20E

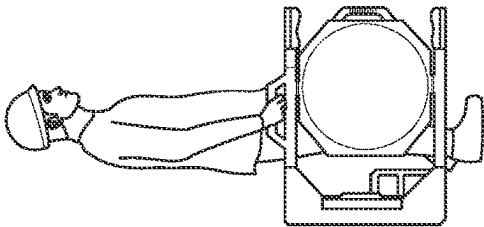


FIG. 21B

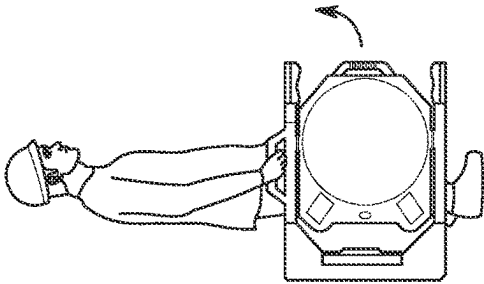


FIG. 21C

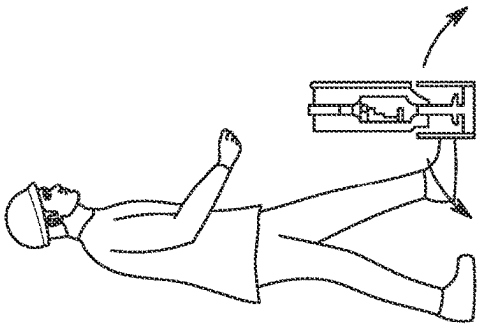


FIG. 21D

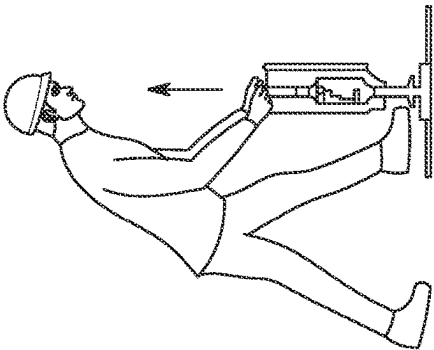


FIG. 21E

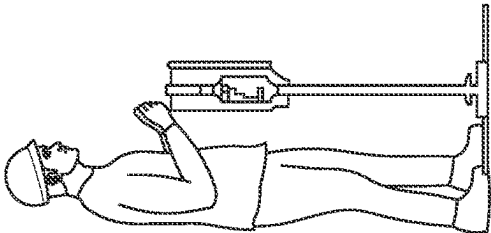


FIG. 21F

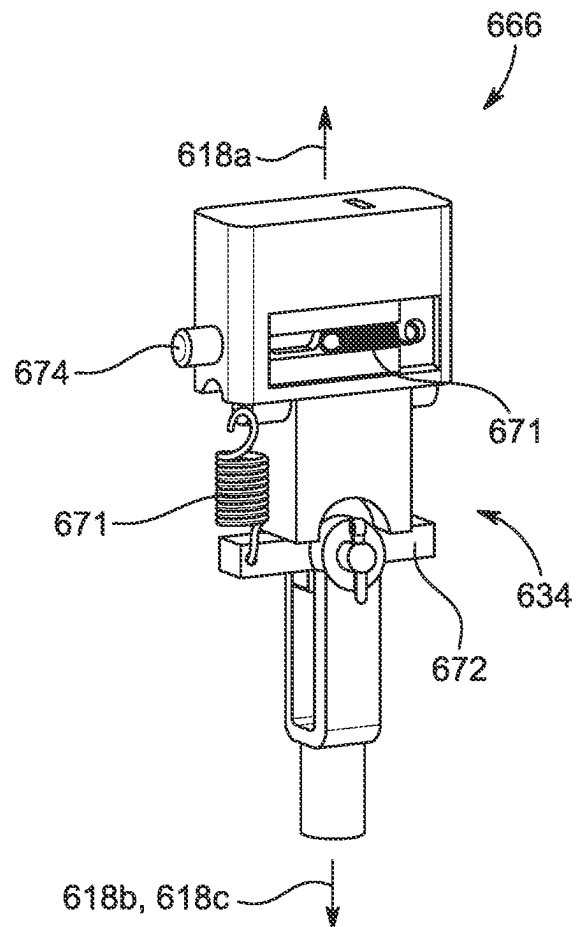


FIG. 21G

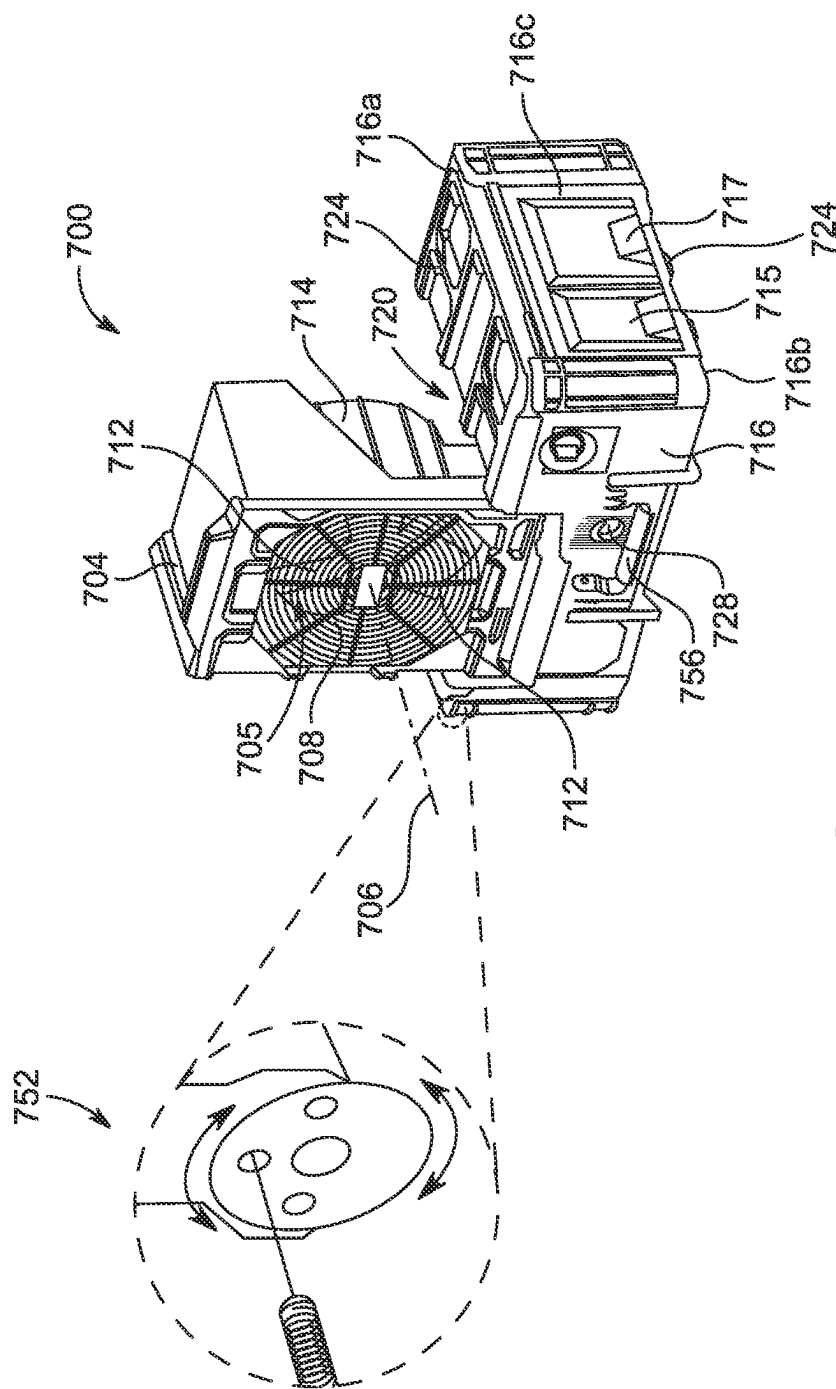


FIG. 22A

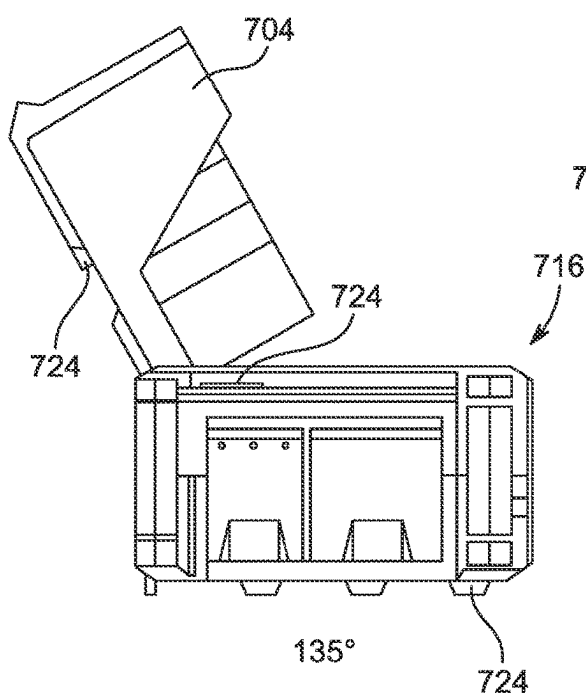


FIG. 22B

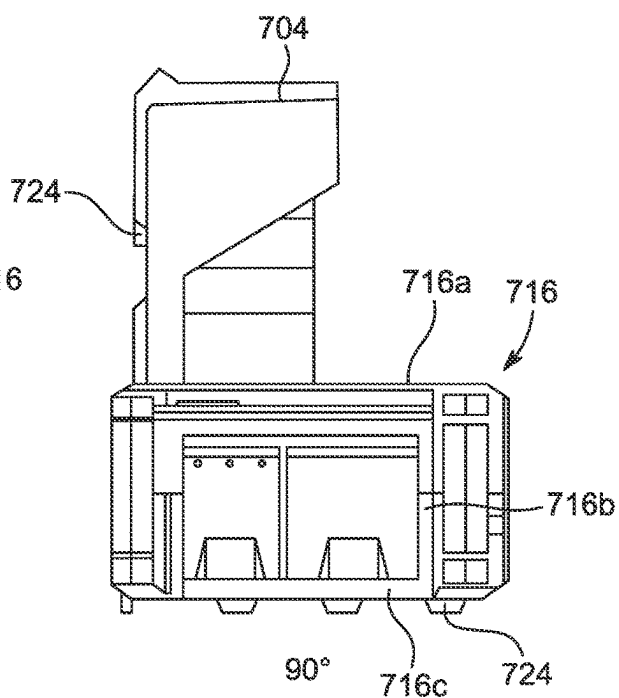


FIG. 22C

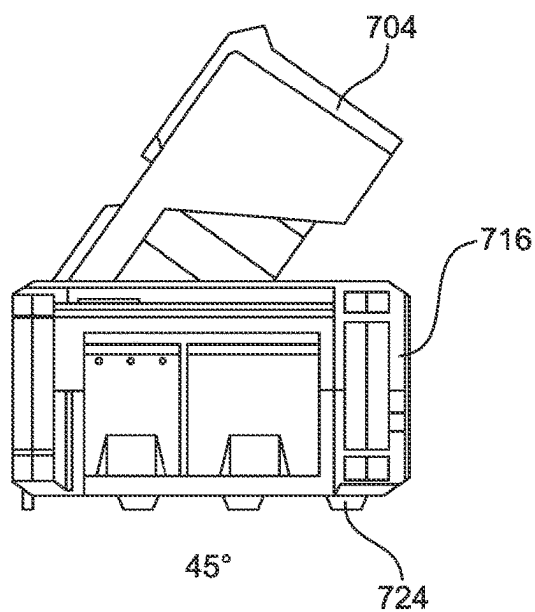


FIG. 22D

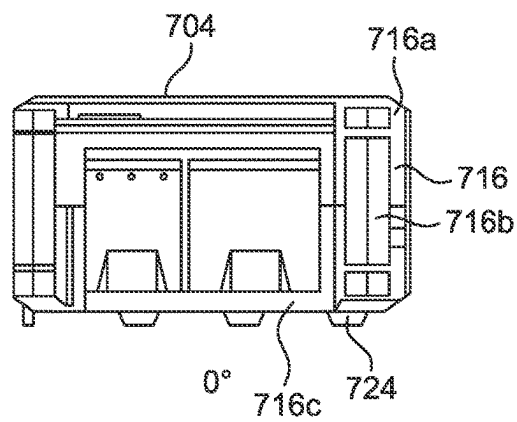


FIG. 22E

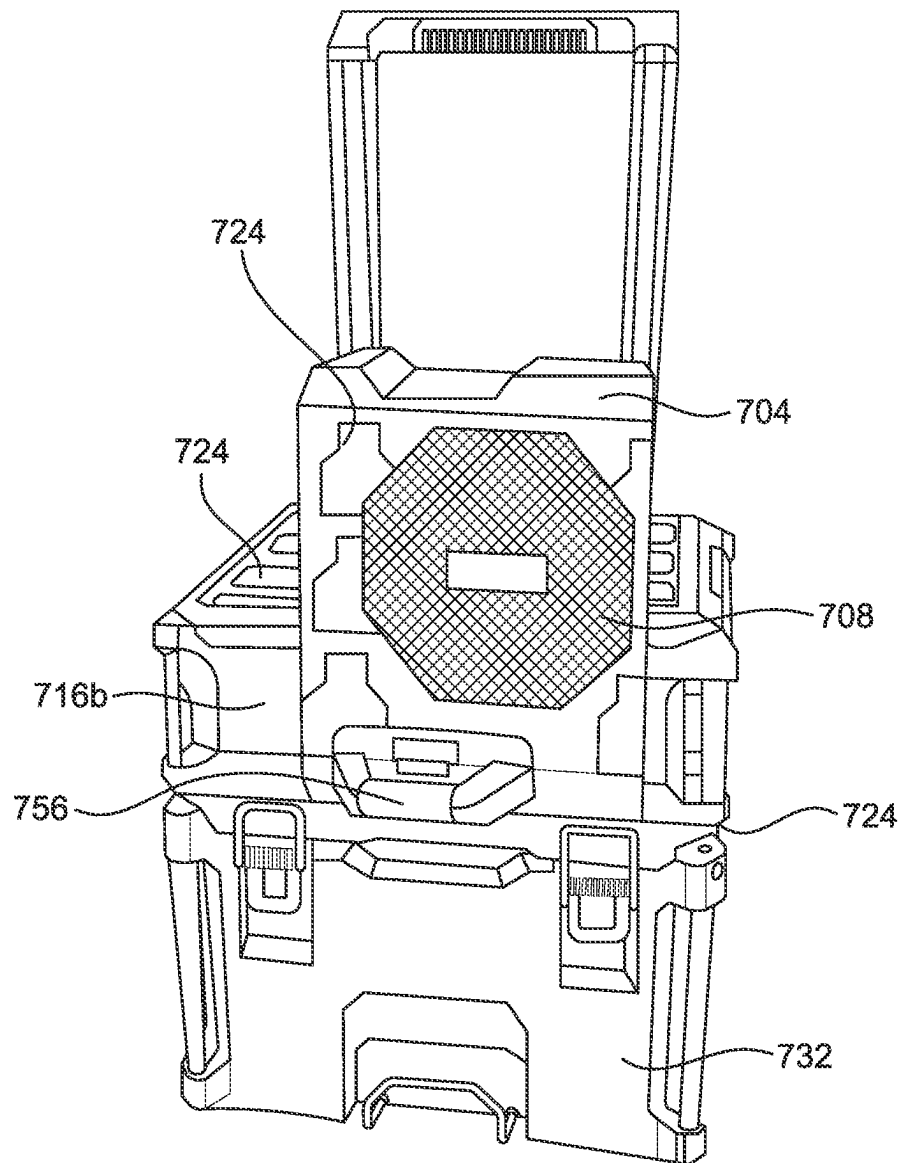


FIG. 22F

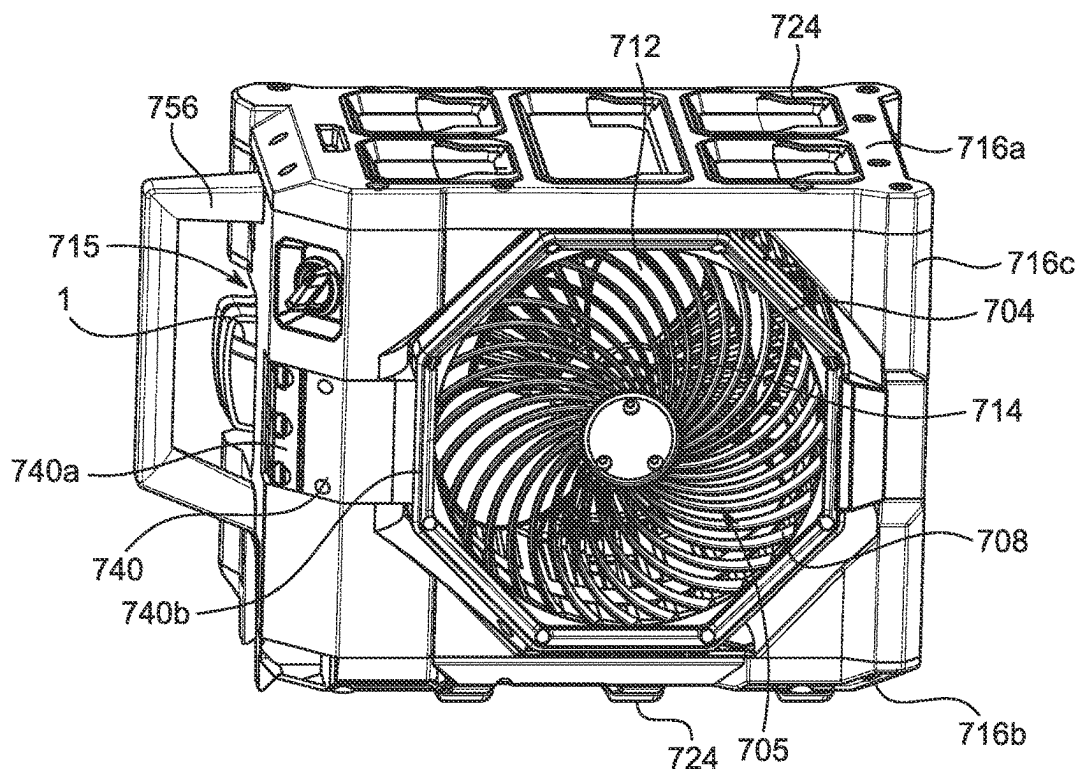


FIG. 23A

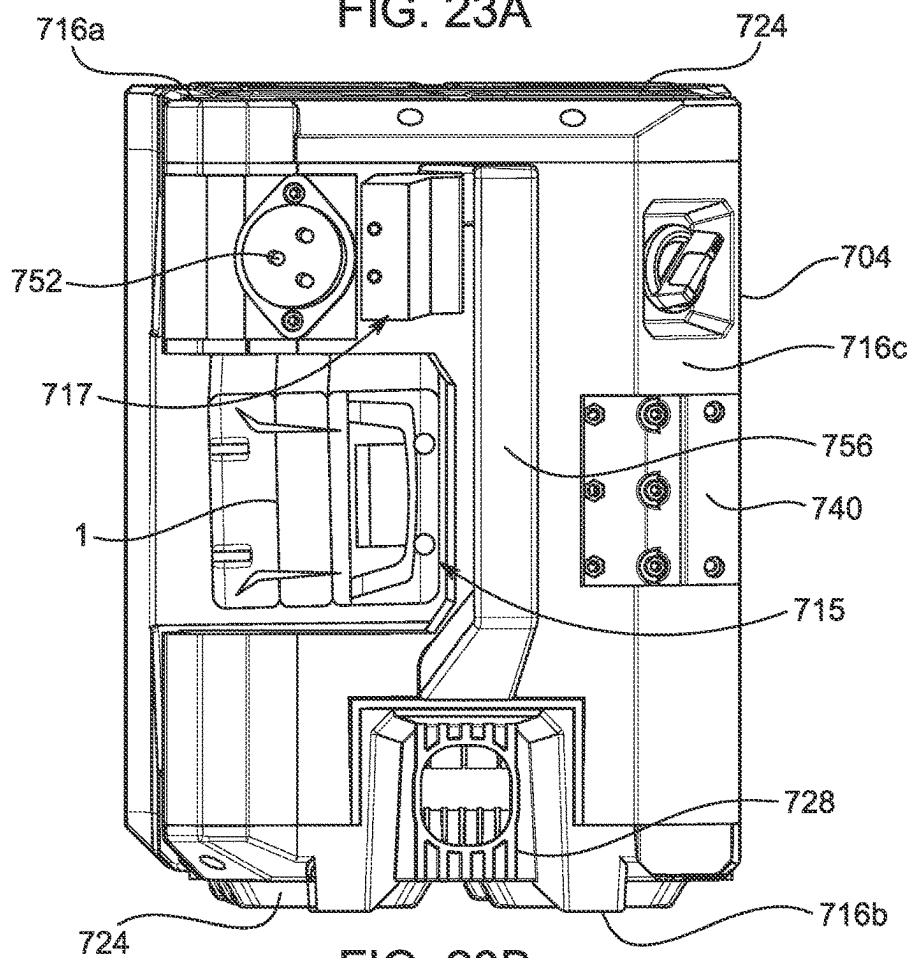


FIG. 23B

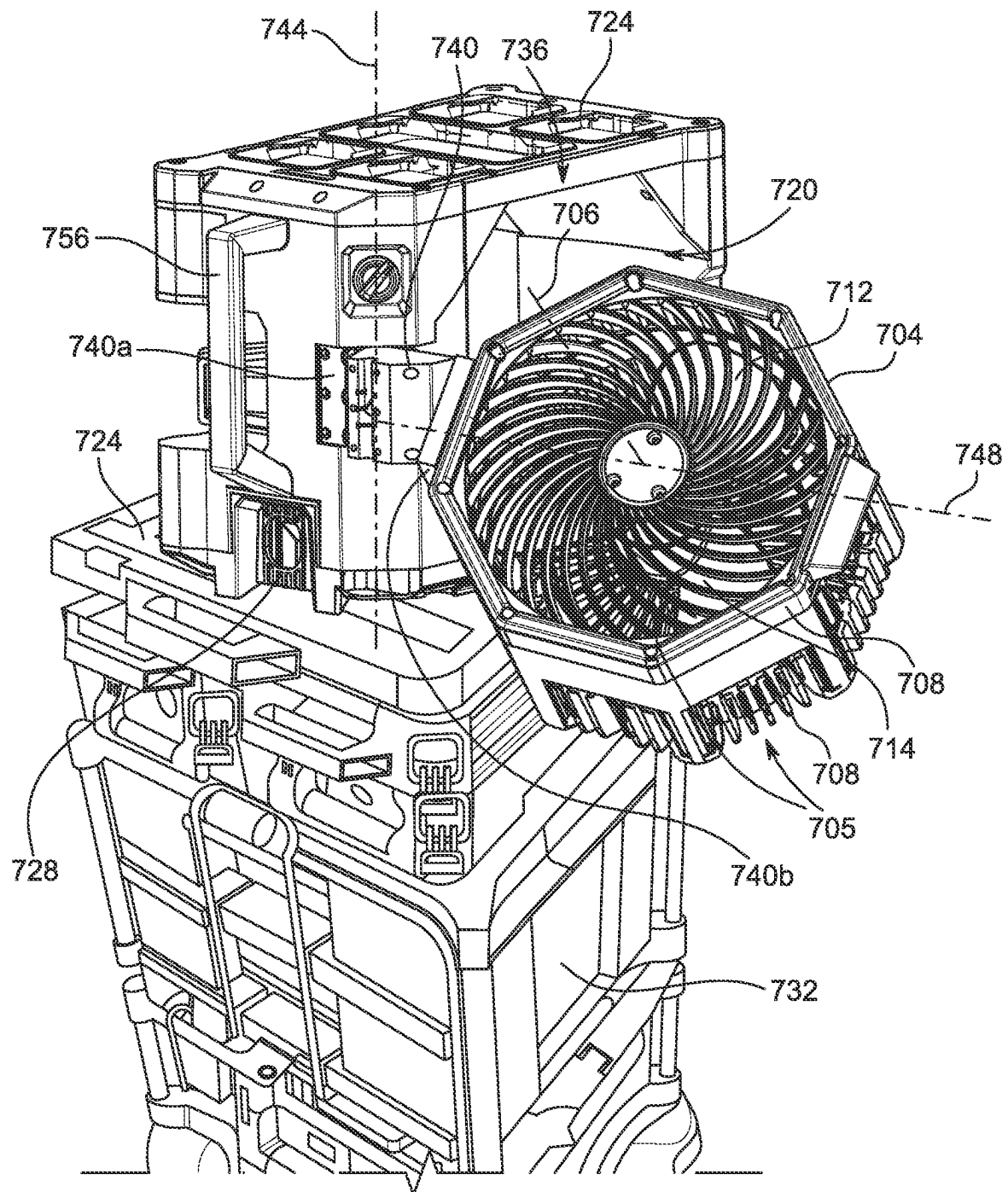


FIG. 23C

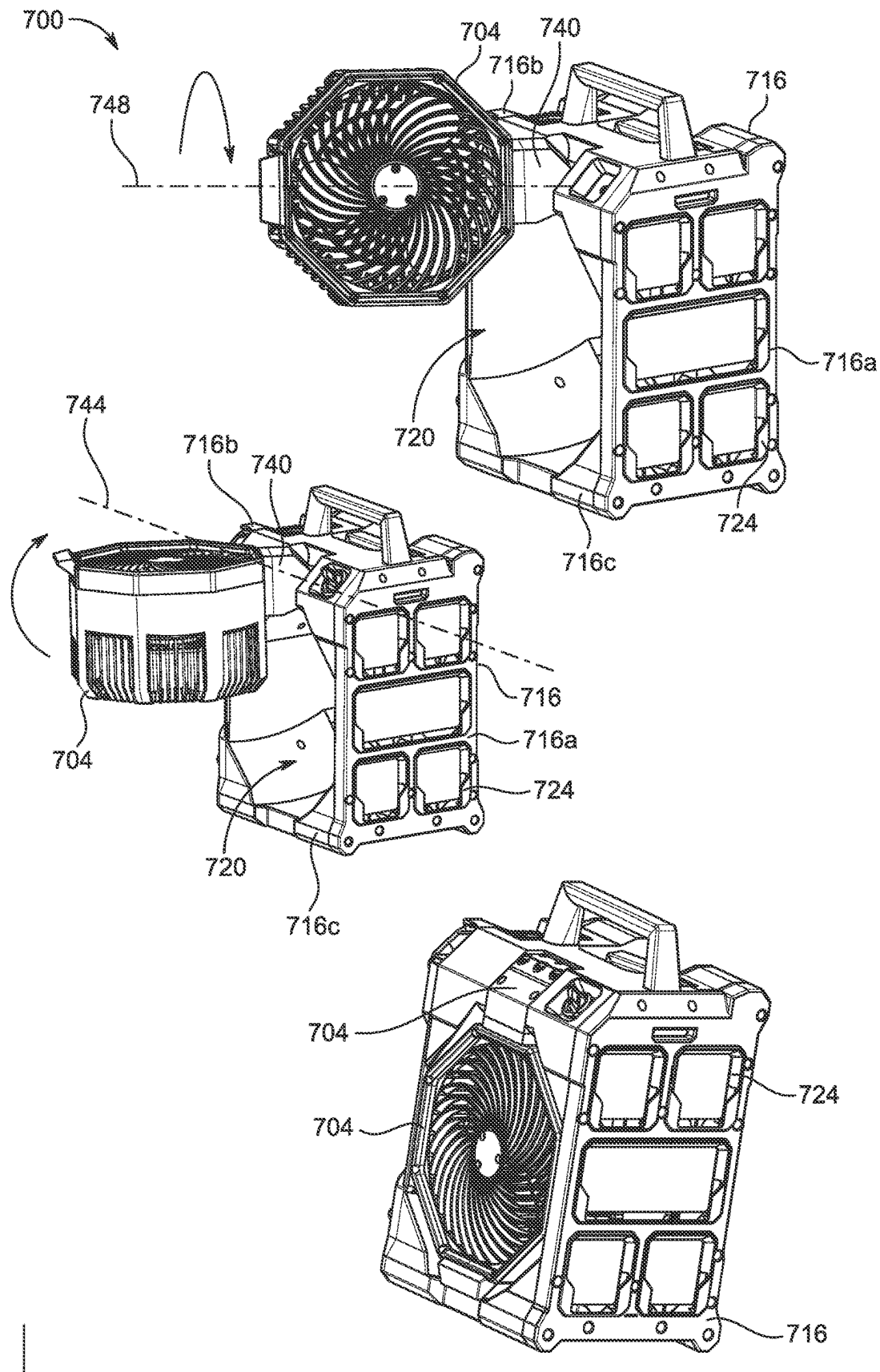


FIG. 23D

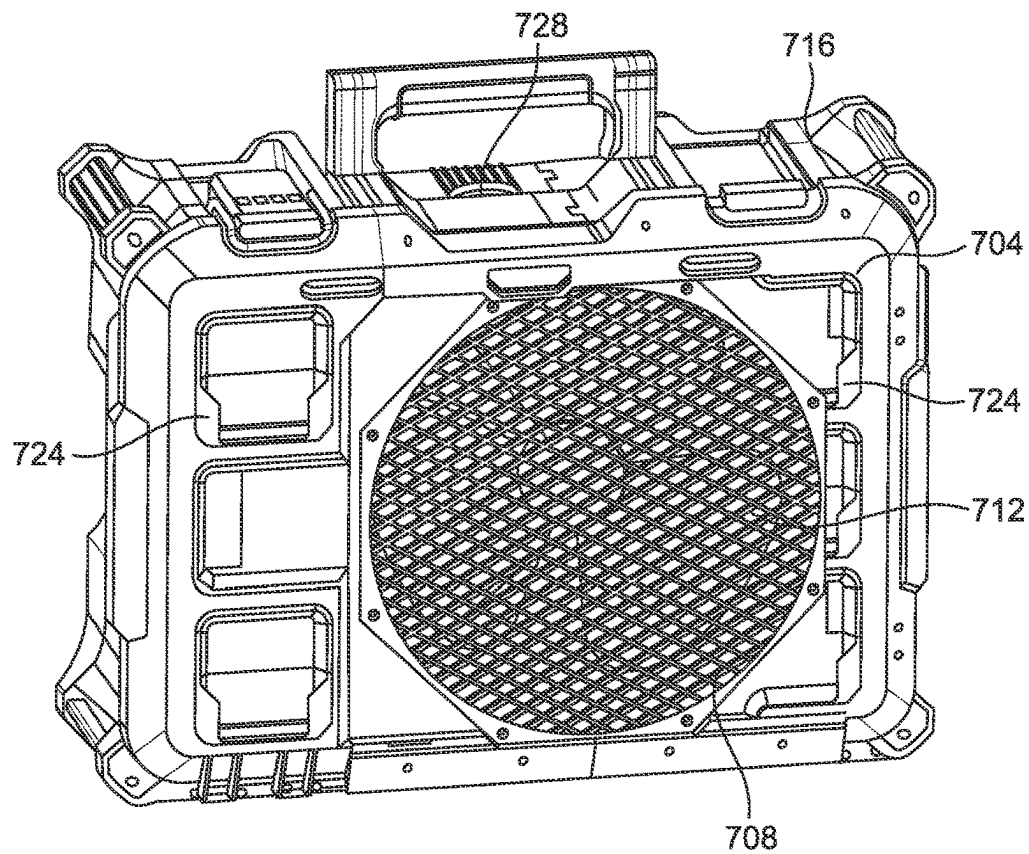


FIG. 24

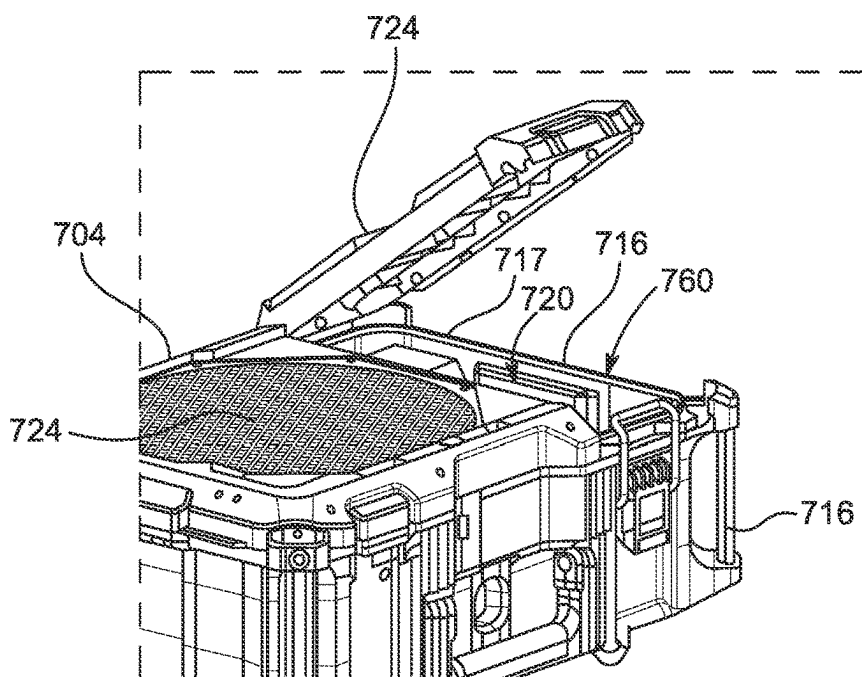


FIG. 25

1

FAN ASSEMBLY**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Patent Application No. 63/160,305, filed Mar. 12, 2021, and to U.S. Provisional Patent Application No. 63/294,935, filed Dec. 30, 2021, the entire contents of which are incorporated by reference herein.

FIELD

The present disclosure relates to cooling systems and, more particularly, to portable fan assemblies configured to be used in a workspace.

SUMMARY

In one embodiment, the invention provides a battery powered box-type fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations. The base includes a stand pivotable relative to the fan housing between a collapsed position, in which the stand is flipped up toward the fan housing, and a deployed position, in which the stand extends away from the fan housing and is configured to contact the support surface.

In another embodiment, the invention provides a battery powered box-type fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface. The base includes a body elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations, a stand movable relative to the body, and a latch coupled to the body and the stand, the latch releasably securing a position of the stand relative to the body, the latch being manually operable to allow movement of the stand relative to the body.

In another embodiment, the invention provides a battery powered box-type fan assembly including a fan housing having a first face, a second face opposite the first face, and a perimeter extending between the first and second faces, the first face, the second face, and the perimeter defining a volume therebetween, the fan housing at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a base pivotally coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface, wherein the fan housing is pivotable relative to the base about a pivot axis that is parallel to the support surface; a battery receptacle positioned on the fan housing and electrically coupled to the motor; and a battery pack removably coupled to the battery receptacle, the battery pack positioned

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within the volume of the fan housing and beneath the pivot axis when coupled to the battery receptacle.

In still another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle configured to receive a removably battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, a second frame segment pivotally coupled to the first frame segment, and a third frame segment pivotally coupled to the second frame segment. The base assembly is configured to be collapsed into a storage configuration in which each of the first, second, and third frame segments are folded onto each other, the base assembly is configured to be expanded to a first deployed configuration in which the fan housing is elevated above the support surface by a first height, and the base assembly is configured to be expanded to a second deployed configuration in which the fan housing is elevated above the support surface by a second height that is different than the first height.

In another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, a second frame segment having a first end and a second end opposite the first end, the first end pivotally coupled to the first frame segment, a third frame segment pivotally coupled to the second end of the second frame segment, and a self-locking latch positioned between the second frame segment and the third frame segment, the self-locking latch biased into a locked position to inhibit relative movement between the second frame segment and the third frame segment.

In yet another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, the first frame segment having a first reduced size gripping portion, a second frame segment pivotally coupled to the first frame segment, the second frame segment having a second reduced size gripping portion, and a third frame segment pivotally coupled to the second frame segment. The base assembly is configured to be collapsed into a storage configuration in which each of the first, second, and third frame segments are folded onto each other, and the base assembly is configured to be expanded to a deployed configuration in which the first, second, and third frame segments are unfolded from one another.

In still another embodiment, the invention provides a fan assembly including a base including walls that at least partially define a storage area therebetween; a fan housing that at least partially defines an airflow chamber, the fan housing moveably coupled to the base between a stored position, in which the fan housing is positioned within the storage area, and a deployed position, in which the fan

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housing is at least partially positioned outside of the storage area; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; a mating interface formed on at least one of the base and the fan housing, the mating interface configured to interact with a complementary mating interface formed on a stackable element of the stackable storage system; and a latch mounted to the base, the latch configured to selectively inhibit relative movement between the fan assembly and the stackable element while the mating interface of the fan assembly interacts with the complementary mating interface of the stackable element.

In another embodiment, the invention provides a fan assembly including a base including a first wall having a first mating interface, the first mating interface configured to interact with another stackable element of the stackable storage system, a second wall opposite the first wall and having a second mating interface, the second mating interface configured to interact with another stackable element of the stackable storage system, and a side wall extending between the first wall and the second wall, the side wall defining an opening to a storage area in the base. The fan assembly further including a fan housing that at least partially defines an airflow chamber, the fan housing being moveably coupled to base between a stored position, in which the fan housing is positioned within the storage area, and a deployed position, in which the fan housing is at least partially extended out of the storage area through the opening; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; and a battery receptacle configured to receive a removable battery pack to power the motor.

In still yet another embodiment, the invention provides a fan assembly including a base including walls that at least partially define a storage area therebetween; a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a coupler including a first end pivotally connected to the base about a first axis, and a second end rotatably connected to the fan housing about a second axis that is transverse relative to the first axis. The fan housing is configured to be selectively pivoted relative to the base about the first axis to move the fan housing into and out of the storage area, and the fan housing is configured to be selectively rotated relative to the base about the second axis to change a direction of airflow generated by the plurality of fan blades.

Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a portable fan assembly, according to one embodiment.

FIG. 2 is a perspective view of the portable fan assembly of FIG. 1, illustrating a fan head in a rotated position.

FIG. 3 is a perspective view of the portable fan assembly of FIG. 1 illustrating a fan head in a rotated position of about 180 degrees.

FIG. 4 is a perspective view of a portable cooling system including one or more fan assemblies, according to another embodiment.

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FIG. 5 is a side perspective view of a single fan assembly of FIG. 4 supported on different support faces.

FIG. 6 is a side perspective view of the portable cooling system of FIG. 4, illustrating multiple fan assemblies in a stacked configuration.

FIG. 7 is a perspective view of a portable fan assembly, according to another embodiment.

FIG. 8 is a side view of the portable fan assembly of FIG. 7 in multiple configurations.

FIG. 9 is a perspective view of the portable fan assembly of FIG. 7 in a standing configuration.

FIG. 10 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the fan assembly in a standing configuration and a hanging configuration.

FIG. 11 includes perspective views of the portable fan assembly of FIG. 10, illustrating the fan assembly in a collapsed configuration.

FIG. 12 is another perspective view of the portable fan assembly of FIG. 10 in the collapsed configuration.

FIG. 13 includes perspective views of the portable fan assembly of FIG. 10, illustrating the fan assembly in the standing configuration and the hanging configuration.

FIG. 14 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the portable fan assembly in multiple extended positions.

FIG. 15 is perspective view of the portable fan assembly of FIG. 14, illustrating the portable fan assembly in a collapsed position.

FIG. 16 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the portable fan assembly in multiple configurations.

FIG. 17 includes perspective views of the portable fan assembly of FIG. 16, illustrating the portable fan assembly in example applications.

FIG. 18A is a perspective view of the portable fan assembly of FIG. 16, illustrating an alternate base assembly useable with the portable fan assembly.

FIG. 18B is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the alternate base in a flipped-down configuration.

FIG. 18C is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the alternate base in a flipped-up configuration.

FIG. 18D is a perspective view of FIG. 18C.

FIG. 18E is a side perspective view of the portable fan assembly of FIG. 18A, illustrating a fan housing in a first orientation.

FIG. 18F is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the fan housing in a second orientation.

FIG. 18G is a perspective view of the portable fan assembly of FIG. 18B, illustrating a filter.

FIG. 19A is a perspective view of a portion of the portable fan assembly of FIG. 18B.

FIG. 19B is an isolated perspective view of a latch portion of the portable fan assembly of FIG. 19A, illustrating a latch in a locked position.

FIG. 19C is an isolated perspective view of the latch portion of the portable fan assembly of FIG. 19A, illustrating the latch in an unlocked position.

FIG. 19D is a perspective view of the portable fan assembly of FIG. 19A, illustrating the base in an intermediate position.

FIG. 19E is a perspective view of the portable fan assembly of FIG. 19A, illustrating the base locked in a flipped-down position.

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FIG. 20A is a perspective view of a portable fan assembly, according to another embodiment.

FIG. 20B is a perspective view of a self-locking latch of the portable fan assembly of FIG. 20A.

FIG. 20C is a side view of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in a collapsed configuration.

FIG. 20D includes side perspective views of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in the collapsed configuration and in a deployed configuration.

FIG. 20E includes perspective views of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in example deployed configurations.

FIG. 21A is a perspective view of a portable fan assembly, according to another embodiment.

FIG. 21B is a perspective view of a portable fan assembly of FIG. 21A, illustrating the portable fan assembly being carried in a collapsed configuration.

FIG. 21C is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly being carried in the collapsed configuration, indicating a direction of rotation for deployment of the portable fan assembly.

FIG. 21D is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly in the collapsed configuration, indicating a direction of rotation for deployment of a base of the portable fan assembly.

FIG. 21E is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly with the base deployed, indicating a direction of deployment for a fan housing.

FIG. 21F is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly in a deployed configuration.

FIG. 21G is a perspective view of the portable fan assembly of FIG. 21A, illustrating a joint of the portable fan assembly.

FIG. 22A is a perspective view of a fan assembly, according to another embodiment.

FIG. 22B is a perspective view of the fan assembly of FIG. 22A in one deployed configuration.

FIG. 22C is a perspective view of the fan assembly of FIG. 22A in another deployed configuration.

FIG. 22D is a perspective view of the fan assembly of FIG. 22A in still another deployed configuration.

FIG. 22E is a perspective view of the fan assembly of FIG. 22A in a collapsed configuration.

FIG. 22F is a perspective view of the fan assembly of FIG. 22A, illustrating the fan assembly stacked on top of another stacking element.

FIG. 23A is a perspective view of a fan assembly, according to another embodiment.

FIG. 23B is another perspective view of the fan assembly of FIG. 23A.

FIG. 23C is a perspective view of the fan assembly of FIG. 23A, illustrating the fan assembly stacked on top of another stacking element.

FIG. 23D includes perspective views illustrating the fan assembly of FIG. 23A in a collapsed configuration and multiple deployed configurations.

FIG. 24 is a perspective view of a fan assembly, according to another embodiment.

FIG. 25 is a perspective view of the fan assembly of FIG. 24, illustrating an open storage compartment on the fan assembly.

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Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Use of “including” and “comprising” and variations thereof as used herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Use of “consisting of” and variations thereof as used herein is meant to encompass only the items listed thereafter and equivalents thereof. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

DETAILED DESCRIPTION

With brief reference to FIGS. 1-24, each embodiment may be powered by a battery pack 1, also referred to as a removable battery pack. The battery pack 1 may be a power tool battery pack generally used to power a power tool, such as an electric drill, an electric saw, and the like (e.g., an 18 volt rechargeable battery pack, or an M18 REDLITHIUM battery pack sold by Milwaukee Electric Tool Corporation). The battery pack 1 may include lithium ion (Li-ion) cells. In alternate embodiments, the battery packs may be of a different chemistry (e.g., nickel-cadmium (NiCa or NiCad), nickel-hydride, and the like). The battery pack 1 may be an 18 volt battery pack, a 4 volt battery pack, a 28 volt battery pack, a 40 volt battery pack, or a battery pack of any other voltage such that the capacity of the battery pack may vary. The battery pack may further include an indicator to display the current state of charge of the battery pack and/or other characteristics of the battery pack.

FIGS. 1-3 illustrate a portable cooling system or fan assembly 10 according to one embodiment. The fan assembly 10 is configured to provide airflow to an area such as a workspace, a workshop, a jobsite, or the like. The fan assembly 10 may include a main body 14 that supports a user interface 18 and a handle 22 configured to accommodate carrying the fan assembly 10. The user interface 18 may be a button, a switch, a toggle, or another selector that allows a user to interact and thus control an operation of the fan assembly 10. In the illustrated embodiment, the fan assembly 10 has an overall rectangular shape, although other polygonal shapes are considered. In some embodiments, the fan assembly 10 may be generally curved and/or oblong.

As illustrated in FIG. 1, the main body 14 is generally cuboid and includes at least four support faces 26A, 26B, 26C, 26D upon which the fan assembly 10 may be supported when, for example, the fan assembly 10 is set on a surface such as a floor surface, table top, or the like. As illustrated in FIGS. 2 and 3, the main body 14 may alternatively have a narrowed or thinned width, thickness, or the like. The fan assembly 10 further includes a first fan head 30A and a second fan head 30B. The fan heads 30A, 30B may each be rotatably coupled to opposing faces 26E, 26F of the main body 14 and rotatable relative to the main body 14 about an axis A-A. The first fan head 30A may support a first fan 34A, such as a blade(s), impeller, or the like rotatably supported therein. The second fan head 30B may support a second fan 34B, such as a blade(s), impeller, or the like rotatably

supported therein. The first fan 34A and the second fan 34B may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

Each fan head 30A, 30B may also be polygonal shaped. In the illustrated embodiment, the main body 14 and fan heads 30A, 30B are similarly shaped and may have similar dimensions. Namely, the support faces 26A, 26B, 26C, 26D of the main body 14 may selectively align with faces 32 of the fan heads 30A, 30B to give the fan assembly 10 a generally rectangular shape. In the illustrated embodiment, each fan head 30A, 30B is generally cuboid.

As best shown in FIG. 2, the fan heads 30A, 30B are rotatable about the axis A-A relative to one another, as well as relative to the main body 14. Each fan head 30A, 30B may further support a shroud 38 and a grille 42. In one example construction of the fan assembly 10, the blades of the first fan 34A and the second fan 34B are rotated to generate an airflow 46 generally entering the respective fan heads 30A, 30B through a rear portion and along the shrouds 38 and exiting through the grilles 42.

As further shown in FIG. 2, a direction of airflow 46 may be selectively changed by rotating one of the fan heads 30A, 30B or by rotating the fan assembly 10. In some scenarios, as shown in FIG. 3, the direction of the airflow 46 generated by each fan head 30A, 30B may be relatively offset by about 180 degrees. In other scenarios, the direction of the airflow 46 generated by each fan head 30A, 30B may be relatively offset by about another about between 0 degrees and 180 degrees, including 0 degrees.

As even further shown in FIG. 2, the fan assembly 10 may be supported on any face 32 of the fan heads 30A, 30B including a face 32 supporting the grille 42, a face 32 intersecting the axis A-A, or another face 32. Each fan head 30A, 30B may further support one or more auxiliary handles 50. The auxiliary handles 50, similar to the handle 22, may be configured to accommodate carrying the fan assembly 10. The user interface 18 may be operable to select a rotational speed and/or rotational direction of the blades of the fans 34A, 34B. In the illustrated embodiment, the user interface 18 is operable to selectively control both of the fans 34A, 34B together, or each of the fans 34A, 34B independently.

FIGS. 4-6 illustrate a portable cooling system 100 including one or more fan assemblies 104 according to another embodiment. The fan assembly 104 supports a fan 110 rotatably mounted therein. The fan 110 is configured to generate an airflow 114 through a fan body or shroud 118 and out a grille 122 positioned on a face of the fan assembly 104. As illustrated in FIG. 5, the fan assembly 104 may be formed of opposing first and second support faces 124A, 124B and opposing third and fourth support faces 128A, 128B. The support faces 124A . . . 128B are generally not faces that support the grille 122, but the fan assembly 104 could still be supported by one of such faces.

The first and second support faces 124A, 124B may be angled relative to the third and fourth support faces 128A, 128B such that a user may select an angle of airflow 114 direction by altering which support face 124A . . . 128B the fan assembly 104 is supported on. For example, when the fan assembly 104 is supported on the third or fourth support face 128A, 128B, airflow 114 may be generally parallel to a surface 132 upon which the fan assembly 104 is supported. When the fan assembly 104 is supported on the first support faces 124A and/or the second support faces 124B, airflow 114 may be generally offset relative the surface 132 upon which the fan assembly 104 is supported. In one embodiment, the airflow 114 is offset relative the surface 132 by

about 4.5 degrees when the fan assembly 104 is supported on the first support faces 124A or the second support faces 124B.

The portable cooling system 100 may include a plurality of fan assemblies 104 that are stackable relative one another at an interface 142. The interface 142 may include complimentary mating features 146 formed on opposing faces of adjacently stacked fan assemblies 104. The mating features 146 allow the fan assemblies 104 to be stacked and retained in a stable stacked configuration.

Each fan assembly 104 may further include a filter slot for replaceably receiving a filter, such as a standard 20 inch furnace filter. The fan assemblies 104 may receive power from the battery pack 1 and supply an AC or DC output through a ground fault circuit interrupting (GFCI) 120V outlet 152. Each fan assembly 104 may even further include a variable speed user interface 156. In some embodiments, each fan assembly 104 may have an overall square dimension of approximately twenty inches.

FIGS. 7-9 illustrate a fan assembly 200, such as a portable fan assembly, according to another embodiment. The fan assembly 200 may include a shroud or housing 204, a grille 208, and a fan 212 rotatably supported within the housing 204. Legs 216 may be rotatably coupled to the fan assembly 200 to support the fan assembly 200 in multiple configurations. In the illustrated embodiment, the legs 216 are rotatably coupled to the housing 204. The fan 212 may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

As shown in FIG. 8, the legs 216 allow the fan assembly 200 to be operated and/or stored in multiple configurations including an elevated and tilted configuration 220, an elevated and level configuration 224, a first low height tilted configuration 228 in which the fan assembly 200 is supported on a surface by a portion of either leg 216 and the housing 204, a second low height tilted or level configuration 232 in which the fan assembly 200 is supported on the surface solely by either leg 216, and a collapsed configuration 240. A height of the fan assembly 200 in any elevated configuration (e.g., elevated and level configuration 224) may be similar to that of a user (FIG. 9). The legs 216 further allow for general universal adjustment of the fan assembly 200 to thereby direct airflow from the fan 212 in a desired direction.

Rotation of the legs 216 relative the housing 204 may be locked in a desired position via leg locks 244. The fan assembly 200 may further include handles 248 for accommodating carrying the fan assembly 200, one or more user interfaces 252 for controlling an operational output of the fan assembly 200, one or more battery packs 1, a filter (e.g., standard 20x20x1 inch air filter), and one or more rotatable ground engaging elements 256 (e.g., wheels, casters, etc.). One or both of the handles 248 may be a flexible material. One or both of the handles 248 may also be rigid and selectively wrapped with a cord, tape, or the like.

FIGS. 10-13 illustrate a fan assembly 300, such as a portable fan assembly, according to another embodiment. The fan assembly 300 may include a shroud or housing 304, a grille 308, and a fan 312 (e.g., blades) rotatably supported within the housing 304. One or more support bars 316 may be rotatably coupled to the fan assembly 300 to support the fan assembly 300 in multiple configurations. In the illustrated embodiment, the support bar 316 is rotatably coupled to the housing 304. The fan 312 may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

As shown in FIGS. 10 and 13, the support bar 316 allows the fan assembly 300 to be operated and/or stored in multiple configurations including an standing configuration 320 in which the fan assembly 300 is supported on a surface by a portion of the support bar 316 and the housing 304, a collapsed configuration 340 (FIGS. 11 and 12), and a hanging configuration 344 in which the fan assembly 300 may be hung or otherwise supported on a structure by links or hooks 346 formed on the support bar 316 and extending therefrom. In one example, the hooks 346 may support the fan assembly 300 in the hanging configuration in which the hooks 346 at least partially wrap around a beam or plank such as a wooden stud. The hooks 346 may also be rotatably coupled to the support bar 316 such that the hooks 346 may be rotated relative the support bar 316 between collapsed (FIGS. 11 and 12) and extended positions.

The fan assembly 300 may further include handles 348 for accommodating carrying the fan assembly 300, one or more user interfaces 352 for controlling an operational output of the fan assembly 300, and one or more battery packs. In one example construction, the fan assembly 300 may have overall dimensions of 16 inches tall (H), 13.5 inches wide (W), and 3.5 inches deep (D).

FIGS. 14 and 15 illustrate a fan assembly 400, such as a portable fan assembly, according to another embodiment. The fan assembly 400 may include a shroud or housing 404, a grille 408, and a fan (e.g., blades) 412 rotatably supported within the housing 404. The housing 404 may be adjustably supported by a base 416 having a support beam 418. The base 416 may include telescoping members 420 that accommodate raising and lowering of the housing 404 relative the support beam 418. In the illustrated embodiment, the support beam 418 may be rotatably or otherwise adjustably attached to the telescoping members 420 between a collapsed position (FIG. 15) and an extended position in which the support beam 418 accommodates standing of the fan assembly 400 on a surface. The fan 412 may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

In one example construction, the base 416 accommodates height adjustment of the housing 404 to different heights between about twenty inches tall and about fifty inches tall, including twenty inches and fifty inches. In another example construction, the base 416 accommodates height adjustment of the housing 404 to heights above a typical workspace fan such as higher than the height of an average user (e.g., 5 feet 8 inches to 7 feet).

The base 416 may further include a pedal 424 that may be operated by a user to selectively lock and unlock the telescoping members 420 relative to the support beam 418. The pedal 424 may be a foot pedal 424 that is configured to be pressed by a foot of a user. For example, the user may step on the pedal 424 to thereby release the telescoping members 420. The user may then lift the housing 404 to a desired height while maintaining pressure on the pedal 424. Once the user releases the pedal 424, the telescoping members 420 may once again be constrained from movement.

In some embodiments the telescoping members are lift actuated by a spring, hydraulic, plunger, or another type of actuator. In some embodiments, the pedal is biased to a locked position. The fan assembly 400 may further include handles 448 for accommodating carriage of the fan assembly 400, one or more user interfaces 452 (e.g., knobs, buttons, switches, etc.) for controlling an operational output of the fan assembly 400, and one or more battery packs 1 configured to power the motor.

FIGS. 16-19E illustrate a fan assembly 500, such as a portable and battery powered box-type fan assembly, according to another embodiment. The fan assembly 500 includes a fan housing 504, a grille 508, a fan 512 (such as fan impeller, a blade, or a plurality of fan blades) rotatably supported within the fan housing 504, and a base 516. The base 516 is configured to be placed on a support surface (e.g., floor surface, table top, windowsill, etc.) and elevate the fan housing 504 above the support surface to not disturb dust or debris below. The fan housing 504 is rotatably coupled to the base 516 such that the base 516 is operable to support the fan housing 504 in multiple directions, orientations, and/or configurations.

The fan 512 may be rotated by a motor 514 that can be powered by a DC power source (e.g., a power tool battery pack), such as the removable battery pack 1, or an AC power source (e.g., a wall outlet). The fan 512 is positioned within an airflow chamber 505 at least partially defined by the fan housing 504. In the illustrated embodiment, the fan 512 is rotated by the motor 514 to generate airflow through the airflow chamber 505. The fan 512 may be rotated by the motor 514 about a rotation axis 506.

The fan assembly 500 further includes a battery receptacle 515 that can receive the removable battery pack 1. In the illustrated embodiment, the battery receptacle 515 is positioned on the fan housing 504. The fan housing 504 may further support an AC power inlet 517 that can provide charging power to the removable battery pack 1 that is received in the battery receptacle 515. In some embodiments, the AC power inlet 517 is positioned on another part of the fan assembly 500, such as the base 516. In either position, the AC power inlet 517 is electrically coupled to the motor 514, to the battery receptacle 515, or to both, such that the motor 514 could be operated on power provided from the battery pack 1 and/or from the AC power inlet 517. An operational output (e.g., speed) of the motor 514 may be controlled by a selector 513, such as a rotational knob, a switch, a button, and/or the like.

Referring to FIG. 16, the base 516, or frame, includes a stand 518 that supports stabilizing legs 520 to keep the fan housing 504 upright during use. The stabilizing legs 520 fold out or flip-down to provide extra stability in a deployed position, and fold in or flip-up for slender storage and transport in a collapsed position. In the deployed positions, the stabilizing legs 520 are configured to contact the support surface. As shown in FIG. 16, the stabilizing legs 520 may be coupled to the fan housing 504 by hinges 522 such that the stabilizing legs 520 rotate 90 degrees between the deployed and collapsed positions. Brackets 524 may also lock the stabilizing legs 520 in place when deployed. In other embodiments, as illustrated in FIG. 17, the brackets may be omitted. Additionally or alternatively, the hinges 522 may each include a one-way locking detent latch 526, as shown in FIG. 19A and described below. In the illustrated embodiment, a pair of stabilizing legs 520 are positioned on each side of the fan housing 504. In other embodiments, stabilizing legs 520 are positioned on a single side of the fan housing 504.

As shown in FIGS. 16, 18E, and 18F, the fan housing 504 is pivotal (e.g., rotatable) relative to the base 516 about a pivot axis 507 that is parallel to the support surface and perpendicular to the rotation axis 506. The stand 518 may include a body 536 and one or more (e.g., two) arms 527 connected by the body 536. In the illustrated embodiment, the body 536 is elongated along a width of the fan housing 510. The arms 527 may extend upwardly from the body 536 and on diametrically opposite sides of the airflow chamber

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505 to support the fan housing 504 through a plurality of directions. For example, the fan housing 504 may rotate to adjust an angle of the fan 512 to alter a direction of the airflow. In some embodiments, the fan housing 504 and/or base 516 may have hard stops 519 (FIGS. 18E and 18F) to inhibit over-rotation relative to each other. For example, the fan housing 504/base 516 may have a first hard stop 519 at a 20 degree downward tilt (FIG. 18E) and a second hard stop 519 at a 50 degree or 90 degree upward tilt (FIG. 18F).

The fan housing 504 may further include a handle 528. The handle 528 may fold down into the fan housing 504. As illustrated in FIG. 18G, the fan housing 504 may also include retainers 530, such as clips, magnets, tracks, and/or the like, located behind the fan 512. The retainers 530 may allow a standard filter (e.g., a 20 inch×20 inch×1 inch filter) to be releasably held behind the fan 512. In the illustrated embodiment, the filter may be placed adjacent an inlet of the airflow chamber 505.

As shown in FIG. 17, the fan assembly 500 may include mounting features 532 to help mount the fan assembly 500 in a window opening. For example, the fan assembly 500 may include a bungee cord 533 or other suitable cord or cable. Additionally or alternatively, the fan assembly 500 may include a hook 534 or other suitable projection. While mounted in a window opening, a front pair of the stabilizing legs 520 and/or a rear pair of the stabilizing legs 520 may be flipped-up. For example, the rear pair of stabilizing legs 520 may be flipped up, and the fan assembly 500 may be at least partially supported by a frame of the window or another portion of the window. In one instance, the fan assembly 500 could be leaned against a screen or frame within an opening window.

Referring now to FIGS. 18A-19E, the fan housing 504 may be supported by an alternate base 516a that is similar to the base 516 illustrated in FIGS. 16 and 17. The alternate base 516a is coupled to the fan housing 504 and is placed on the support surface and includes a body 536 that elevates the fan housing 504 above the support surface. The alternate base 516a and body 536 are also operable to support the fan housing 504 in multiple configurations, orientations, directions, etc. The alternate base 516a further includes a stand 540 movable relative to the body 536 and the one-way locking detent latch 526, which is coupled to the body 536 and to the stand 540. The detent latch 526 releasably secures a position of the stand 540 relative to the body 536 and is operable (e.g., by a user) to allow movement of the stand 540 relative to the body 536. The detent latch 526 is further positioned adjacent at least one of the hinges 522.

As specifically illustrated in FIGS. 18A-18G, the fan housing 504 has a first face 546, a second face 548 opposite the first face 546, and a perimeter extending between the first face 546 and the second face 548. In other words, the perimeter is made up of the plurality of faces 550 extending between the first face 546 and the second face 548. In the illustrated embodiment, the perimeter is polygonal (e.g., rectangular, square, hexagonal, etc.) but could be circular in other embodiments. In the illustrated embodiment, a perimeter of the airflow chamber 505 is circular, but could be polygonal in other embodiments.

The first face 546, the second face 548, and the plurality of faces 550 (e.g. the perimeter) together define a volume therebetween. In the illustrated embodiment, the battery receptacle 515 is positioned within the volume adjacent the second face 548 of the fan housing 504. In the illustrated embodiment, the first face 546 includes the retainers 530 for the replaceable filter. Referring specifically to FIG. 18A, the battery receptacle 515 is positioned adjacent a vertex of two

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side faces (e.g., faces 550) and below both the rotation axis 506 and the pivot axis 507. The position of the battery receptacle 515 on the fan housing 504 provides additional support to the fan assembly 500 while the battery pack 1 is received in the battery receptacle 515. In some instances, the position of the battery receptacle 515 makes movement of the fan housing 504 easier for the user and resists the fan housing 504 from being flipped over.

In the illustrated embodiment, as illustrated in FIGS. 19B and 19C, the detent latch 526 includes a plurality of ramps 542a each terminated by a shoulder 544a. The stand 540 includes complementary ramps 542b and shoulders 544b that are radially offset relative the ramps 542a and the shoulders 544a such that the detent latch 526 and the stand 540 cooperate with one another in a one-way ratchet manner. As such the detent latch 526 includes a ratchet mechanism that has a series of ramps 542a and shoulders 544a that contact the complementary series of ramps 542b and shoulders 544b formed on the stand 540 to permit ratcheting rotation of the stand 540 relative to the body 536.

As further illustrated in FIGS. 19B and 19C, the stand 540 may be rotatably coupled to the body 536 with the hinge 522, which may be an elongated hinge that extends along the width of the fan housing 504 between the diametrically opposite sides of the fan housing 504. The hinge 522 may also attach a single pair of stabilizing legs 520 together. The detent latch 526 may be biased by a biasing member 554 (e.g., spring) and moveable axially along a portion of the hinge 522. The biasing member 554 may bias the detent latch 526 into a locked position (FIG. 19B) in which the stand 540 is moveable relative the body 536 in a first direction or flip-down direction 556 (arrow in FIG. 19D) but inhibited from flipping up by contact between the shoulders 544a on the detent latch 526 and the shoulders 544b on the stand 540. As the stand 540 moves in the flip-down direction 556, the ramps 542a, 542b engage and slide on one another. As the ramps 542a, 542b slide, the biasing member 554 is engaged and allows the detent latch 526 to slide until the 542a, 542b clear one another and the shoulders 544a, 544b slide into engagement with one another, in which the detent latch 526 will prevent movement of the stand 540 in the opposing flip-up direction.

Once the stand 540 is flipped down so that the stabilizing legs 520 are parallel with the support surface, as illustrated in FIGS. 18B and 19E, the detent latch 526 is actuated, such as by the hand or foot of a user, to separate the shoulders 544a, 544b from one another, and allow the stand 540 to be flipped up (e.g., from the deployed position toward the collapsed position). In some embodiments, the alternate base 516a includes a stand 540 on each side of the fan housing 504, and each stand 540 is moveable independently from the other. In such embodiments, the fan assembly 500 includes a detent latch 526 on each stand 540.

FIGS. 20A-21G illustrate a fan assembly 600, such as a portable fan assembly, according to another embodiment. The fan assembly 600 includes a fan housing 604, a grille 608, a fan 612 rotatably supported within the fan housing 604, and a base assembly or base 616. The fan 612 may be or include an impeller, a blade, a plurality of blades, or the like, and may be rotated by motor 614 powered by a DC power source, such as the battery pack 1, or an AC power source (e.g., a wall outlet). The fan 612 may be supported by the fan housing 604 within an airflow chamber at least partially defined by the fan housing 604. FIGS. 20A-20E illustrate a base 616 useable with the fan housing 604, and FIGS. 21A-21F illustrated an alternate base 616a useable with the fan housing 604 or a slightly modified fan housing.

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The base **616** and the alternate base **616a** are configured to be placed on a support surface (e.g., floor surface, table top, windowsill, etc.) and elevate the fan housing **604** above the support surface to not disturb dust or debris below. The fan housing **604** is rotatably coupled to the base **616** and the alternate base **616a** such that the base **616** and the alternate base **616a** are operable to support the fan housing **604** in multiple directions, orientations, and/or configurations. The fan **612** is positioned within an airflow chamber **605** at least partially defined by the fan housing **604**. In the illustrated embodiment, the fan **612** is rotated by the motor **614** to generate airflow through the airflow chamber **605**. The fan **612** may be rotated by the motor **614** about a rotation axis **606**.

The fan assembly **600** further includes a battery receptacle **615** that can receive the removable battery pack **1**. In the illustrated embodiment of FIGS. **20A-20E**, the battery receptacle **615** is positioned on the fan housing **604**. In the illustrated embodiment of FIGS. **21A-21G**, the battery receptacle **615** is positioned on the alternate base **616a**. The fan housing **604** and/or the alternate base **616a** may further support an AC power inlet **617** that can provide charging power to the removable battery pack **1** that is received in the battery receptacle **615**. In some embodiments, the AC power inlet **617** is positioned on another part of the fan assembly **600**, such as the base **616**. In either position, the AC power inlet **617** is electrically coupled to the motor **614**, to the battery receptacle **615**, or to both, such that the motor **614** could be operated on power provided from the battery pack **1** and/or from the AC power inlet **617**.

The base **616**, or frame, includes a first frame segment **618a** having a first end **619** and an opposite second end **620**, a second frame segment **618b** having a first end **621** and a second end **622**, and a third frame segment **618c** having a first end **623** and a second end **624**. Each of the illustrated frame segments **618a**, **618b**, **618c** includes two parallel, spaced apart beams or members. In other embodiments, each frame segment **618a**, **618b**, **618c** may only include a single beam or member. The first frame segment **618a** can be pivotally coupled to the second frame segment **618b**, and the second frame segment **618b** may be pivotally coupled to the third frame segment **618c**. The first frame segment **618a**, the second frame segment **618b**, and the third frame segment **618c** may be expanded (e.g., unfolded) into a plurality of deployed configurations (FIGS. **20A** and **20E**) and collapsed (e.g., folded, folded down, etc.) into a compact form or storage configuration (FIG. **20C**). The fan housing **604** may rotate or pivot relative to the base **616**. For example, the fan housing **604** may rotate **360** degrees relative to the base **616**. In some embodiments, rotation of the fan housing **604** may be limited to **320** degrees. In the illustrated embodiment, the fan housing **604** is coupled to the first frame segment **618a**, but could be alternately coupled to other portions of the base **616**.

The base **616** may elevate the fan housing **604**, for example, **15** inches, **32** inches, or **42** inches off the ground when in either of the deployed configurations. As illustrated in FIG. **20E**, the base **616** is expandable into a first deployed configuration (left view of FIG. **20E**) in which the first frame segment **618a** is folded on the second frame segment **618b**, the second frame segment **618b** is unfolded from the third frame segment **618c**, and the fan housing **604** is elevated above the support surface by a first height between approximately ten inches and approximately twenty inches (e.g., **15** inches). As also illustrated in FIG. **20E**, the base **616** is expandable into a second deployed configuration (right view of FIG. **20E**) in which the first frame segment **618a** is

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unfolded from the second frame segment **618b**, the second frame segment **618b** is unfolded from the third frame segment **618c**, and the fan housing **604** is elevated above the support surface by a second height between approximately thirty inches and approximately fifty inches (e.g., **32** inches, **42** inches, etc.). In other deployed configurations, the first frame segment **618a**, the second frame segment **618b**, and the third frame segment **618c** may be expanded into different configurations to achieve different heights of the fan housing **604**.

As best illustrated in FIGS. **20A** and **20C**, the first frame segment **618a** may include a first gripping portion **626** and the second frame segment **618b** may include a second gripping portion **630**. The first gripping portion **626** and the second gripping portion **630** may be reduced in size relative to the adjacent portions of the base **616** and may aid in the set-up and adjustment of the fan assembly **600**. The first gripping portion **626** and the second gripping portion **630** may include abrasive or textured surfaces. As shown in FIG. **20C**, the first gripping portion **626** and the second gripping portion **630** may be adjacent one another while the base **616** is collapsed into the storage configuration to provide a relief space for a user to grasp when unfolding or deploying the base **616**. For example, when the first gripping portion **626** and the second gripping portion **630** are adjacent to one another, a user may fit their fingers into the relief space provided to grasp the base **616**.

Referring now to FIG. **20D**, the base **616** further includes a first joint **632** connecting the fan housing **604** and the first frame segment **618a**, a second joint **634** connecting the first frame segment **618a** and the second frame segment **618b**, and a third joint **636** connecting the second frame segment **618b** to the third frame segment **618c**. In the illustrated embodiment, the first joint **632** is positioned between the first end **619** and the second end **620** of the first frame segment **618a**, the second joint **634** is positioned at the first end **623** of the second frame segment **618b**, and the third joint **636** is positioned at the second end **624** of the second frame segment **618b**. In some embodiments, the joints are positioned on other parts of the frame segments.

With continued reference to FIG. **20D**, the second frame segment **618b** is pivotal toward the storage configuration relative to the third frame segment **618c** in a first direction **640**, and the first frame segment **618a** is pivotal toward the storage configuration relative to the second frame segment **618b** in a second direction **644**. In the illustrated embodiment, the first direction **640** and the second direction **644** are opposite to one another, but could be generally similar to one another in other embodiments.

Referring now to FIGS. **20A** and **20B**, the fan assembly **600** further includes a latch or self-locking latch **648** positioned between the second frame segment **618b** and the third frame segment **618c**. In some embodiments, the third joint **636** includes the self-locking latch **648**. In the illustrated embodiment, the self-locking latch **648** is biased into a locked position (left view of FIG. **20B**) in which the self-locking latch **648** inhibits relative movement between the second frame segment **618b** and the third frame segment **618c**. The during deployment of the fan assembly **600**, the second frame segment **618b** is moveable relative to the third frame segment **618c** without the user needing to operate the self-locking latch **648**. The self-locking latch **648** has a ramp **650** and a shoulder **652** that are engaged by a ramp **654** and a shoulder **656** on the first end **623** of the second frame segment **618b**. The ramp **654** of the second frame segment

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618b displaces the ramp **650** of the self-locking latch **648** as the second frame segment **618b** is pivoted into the deployed condition.

Once the ramp **654** of the second frame segment **618b** clears the ramp **650** of the self-locking latch **648**, the shoulder **652** of the self-locking latch **648** bears against the shoulder **656** of the second frame segment **618b** to prevent relative pivoting between the two. While locked, the self-locking latch **648** may be operated (e.g., by the hand or foot of a user) into an unlocked or unlatched positioned (right view of FIG. 20B) to allow relative pivoting between the second frame segment **618b** and the third frame segment **618c**. The fan assembly **600** may even further include a hinge or friction hinge **670** positioned between the first frame segment **618a** and the second frame segment **618b**. The friction hinge **670** is configured to inhibit the first frame segment **618a** and the second frame segment **618b** from swinging (e.g., pivoting) freely from one another. Stated another way, the friction hinge **670** internal pin or slide mechanisms to resist movement withing the second joint **634**. The friction hinge **670** may also be positioned in the first joint **632** and/or the third joint **636**. In general, the self-locking latch **648** or latches and the friction hinge **670** or hinges may keep the base **616** upright and locked while in use.

As illustrate in FIG. 21A, the fan assembly **600** may even further include a handle or carrying handle **664** supported on at least one of the fan housing **604** and the alternate base **616a**. In the illustrated embodiment, the carrying handle **664** is configured to accommodate carrying of the fan assembly **600** in any of the storage configuration, the first deployed configuration, and the second deployed configuration. With further reference to FIG. 21A, the first frame segment **618a**, the second frame segment **618b**, and the third frame segment **618c** of the alternate base **616a** may be telescopically connected to one another such that the second joint **634** and the third joint **636** are sliding joints rather than pivoting joints. In the illustrated embodiment, pivoting or sliding joints may include friction hinges and/or self-locking latched.

In some embodiments, as illustrated in FIG. 21G, the alternate base **616a** may include a slider-pin latch **666** within the second joint **634**. The slider-pin latch **666** includes a spring assembly **671**, a retainer **672**, and a button **674** biased outwardly by the spring assembly **671**. The slider-pin latch **666** is positioned within one of the first frame segment **618a** and the second frame segment **618b**, and the button **674** is positioned to protrude outwardly from the second joint **634** to prevent sliding between the first frame segment **618a** and the second frame segment **618b**. The button **674** may be pressed by a user to rotate the retainer **672** out of preventing engagement between the first frame segment **618a** and the second frame segment **618b** and to allow relative sliding therebetween.

With reference back to FIGS. 21B-21F, the fan assembly **600** may be carried by a user via the carrying handle **664** (FIG. 21B) while in the collapsed position. The user may rotate the fan assembly **600** upward (FIG. 21C) and set on the support surface (FIG. 21D). The user may then place the base **616** or the alternate base **616a** while placing their foot on a portion of the base **616** or the alternate base **616a** (FIG. 21E). The user may then grasp portions of the fan assembly **600**, such as the first gripping portion **626**, the second gripping portion **630**, the first joint **632**, etc. and lift the fan housing **604** relative to the base **616** or the alternate base **616a** to move the fan assembly **600** from the collapsed configuration to the deployed configuration (FIG. 21F). For

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example, the user may press the button **674** or buttons to allow movement between the first frame segment **618a** and the second frame segment **618b** and move the fan assembly **600** toward the deployed configuration. In some embodiments, the fan assembly **600** includes flip-up/flip-down legs **665** that are selectively moveably relative the base **616** (e.g., the third frame segment **618c**) or alternate base **616a** to provide more stability to the fan assembly **600**.

FIGS. 22A-23D illustrate a fan assembly **700**, such as a portable fan assembly, according to another embodiment. The fan assembly **700** includes a fan housing **704**, a grille **708**, a fan **712** such as fan impeller, a blade, or a plurality of fan blades, rotatably supported within the fan housing **704**, and a base **716**. The fan **712** may be powered by a DC power source (e.g., a power tool battery back) or an AC power source (e.g., a wall outlet). The fan assembly **700** also includes an AC pass through.

The fan **712** may be rotated by a motor **714** that can be powered by a DC power source (e.g., a power tool battery back), such as the removable battery pack **1**, or an AC power source (e.g., a wall outlet). The fan **712** is positioned within an airflow chamber **705** at least partially defined by the fan housing **704**. In the illustrated embodiment, the fan **712** is rotated by the motor **714** to generate airflow through the airflow chamber **705**. The fan **712** may be rotated by the motor **714** about a rotation axis **706**.

The fan assembly **700** further includes a battery receptacle **715** that can receive the removable battery pack **1**. In the illustrated embodiment, the battery receptacle **715** is positioned on the fan housing **704**. The base **716** may further support an AC power inlet **717** that can provide charging power to the removable battery pack **1** that is received in the battery receptacle **715**. In some embodiments, the AC power inlet **717** is positioned on another part of the fan assembly **700**, such as the fan housing **704**. In either position, the AC power inlet **717** is electrically coupled to the motor **714**, to the battery receptacle **715**, or to both, such that the motor **714** could be operated on power provided from the battery pack **1** and/or from motor **714**. The fan **712** is positioned within an airflow chamber **705** at least partially defined by the fan housing **704**. In the illustrated embodiment, the fan **712** is rotated by the motor **714** to generate airflow through the airflow chamber **705**. The fan **712** may be rotated by the motor **714** about a rotation axis **706**.

The base **716** may include walls **718** that define a storage area **720** or storage areas. For example, the base **716** may be a toolbox. The base **716** also includes a mating interface **724** configured to interact with other storage units, such as the Milwaukee Tool PACKOUT Modular Storage System. The storage area **720** may also be a space defined in the base **716** for receiving the fan housing **704**. As illustrated in FIGS. 22B-22E and 23D, the fan housing **704** is moveably coupled to the base **716** between a stored position (FIGS. 22E and 23A), in which the fan housing **704** is positioned within the storage area **720**, and a deployed position (FIGS. 22B-22D, 22F, and 23C), in which the fan housing **704** is at least partially positioned outside of the storage area **720**. In the illustrated embodiment, the fan housing **704** may pivot relative to the base **716** between the stored and deployed positions.

In some embodiments, as illustrated in FIG. 22A-22F, **24**, and **25**, the mating interface **724** is formed on the fan housing **704**, such as on the grille **708** at least partially overlaying the airflow chamber **705**, and the base **716**. The fan assembly **700** further includes a latch **728** moveably mounted to the base **716** and configured to selectively inhibit relative movement between the fan assembly **700** and

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another stackable element **732**, such as a toolbox, rolling cart, container, and/or the like while the mating interface **724** of the fan assembly **700** interacts with a complementary mating interface of the stackable element **732**. When the mating interface **724** is formed on the grille **708**, the fan housing **704** is retained in the storage area **720** (e.g., in the stored position) while the stackable element **732** is stacked on top of the fan assembly **700**.

As illustrated in FIGS. **23A-23D**, the mating interface **724** is not formed on the grille **708**, but rather formed on the base **716** such that the fan housing **704** is moveable out of the storage area **720** (e.g., to the deployed position) even while the stackable element **732** is stacked on top of the fan assembly **700**. In other words, when the mating interface **724** is formed on the fan housing **704**, the fan housing **704** is not able to move between the stored position and the deployed position while the stackable element **732** is stacked on top of the fan assembly **700**, and when the mating interface **724** is formed on the base **716**, the fan housing **704** is able to move between the stored position and the deployed position while the stackable element **732** is stacked on top of the fan assembly **700** such that the fan housing **704** is operable to generate airflow in a plurality of directions even when the fan assembly **700** and the stackable element **732** are stacked. The mating interface **724** further supports stacking and interlocking of the fan assembly **700** on top of the stackable element **732**. In some embodiments, the fan assembly **700** can be stacked on top of the stackable element **732** (e.g., on a lid of the stackable element **732**) in a side-by-side arrangement with another fan assembly **700** or another stackable element **732**.

As illustrated in FIGS. **23A-23D**, the base **716** includes a first wall **716a** having a first part of the mating interface **724** configured to interact with the stackable element **732**, a second wall **716b** opposite the first wall **716a** having a second part of the mating interface **724** configured to interact with the stackable element **732**, and a wall or a plurality of walls **716c**, extending between the first wall **716a** and the second wall **716b**, defining an opening **736** to the storage area **720** in the base **716**. In the deployed position, the fan housing **704** is at least partially extended out of the storage area **720** through the opening **736**. In the illustrated embodiment, the plurality of walls **716c** defines an outer perimeter of the base **716**. In some embodiments, the outer perimeter is polygonal. In some embodiments, the outer perimeter is circular. In the illustrated embodiment, the outer perimeter is rectangular, and may even be square. As illustrated in FIG. **23C**, the fan assembly **700** can be stacked on the stackable element **732** in an upright position in which the first wall **716a** and the second wall **716b** are positioned at opposing upper and lower sides of the base **716**, and the plurality of walls **716c** are positioned on sides of the base **716**.

Referring now to FIGS. **23C** and **23D**, the fan assembly **700** further includes a joint or coupler **740** moveably connecting the fan housing **704** to the base **716**. The coupler **740** has a first end **740a** pivotally connected to the base **716** about a first axis **744**, and a second end **740b** rotatably connected to the fan housing **704** about a second axis **748** that is transverse relative to the first axis **744**. The fan housing **704** is configured to be selectively pivoted relative to the base **716** about the first axis **744** to move the fan housing **704** into and out of the storage area **720** and selectively rotated relative to the base **716** about the second axis **748** to change a direction of airflow generated by the fan **712**. In other words, the coupler **740** is moveable to alter both a pitch and a yaw of the fan housing **704** relative the

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base **716**. As best illustrated in FIG. **23C**, both the pitch and the yaw of the fan housing **704** (e.g., first axis **744** and second axis **748**) may be altered while the stackable element **732** is stacked on the fan assembly **700** to move the fan housing **704** between the stored position and the deployed position. As such, the fan housing **704** may move about two degrees of freedom relative to the base **716**.

In some embodiments, as illustrated in FIGS. **22A** and **23B**, a detent mechanism **752** may releasably hold the fan housing **704** in set positions (e.g., 0 degrees, 45 degrees, 90 degrees, and 135 degrees) relative to the base **716**. In other words, the detent mechanism **752** is configured to be set to a predetermined position by a user, where the predetermined position corresponds to an angular position of the fan housing **704** relative to the base **716**. In some embodiments, the fan assembly **700** even further includes a handle **756** configured to accommodate carrying of the fan assembly **700** while stacked with or unstacked from the stackable element **732**. The handle **756** may be positioned on the fan housing **704**, the base **716**, or both.

In some embodiments, as illustrated in FIGS. **24** and **25**, the fan housing **704** may be smaller than the outer perimeter of the fan assembly **700** or base **716**, and the fan assembly **700** may further include a storage compartment **760** separate from the storage area **720**. The storage compartment **760** may hold tools, tool bits, and/or the like. In some embodiments, the battery receptacle **715** and/or AC power inlet **717** are positioned within the storage compartment **760**. The storage compartment **760** may also be waterproof and/or sealable.

Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described. Various features and advantages of the invention are set forth in the following claims.

What is claimed is:

1. A battery powered box-type fan assembly comprising:
 - a fan housing that at least partially defines an airflow chamber;
 - a plurality of fan blades positioned within the airflow chamber;
 - a motor operable to rotate the plurality of fan blades;
 - a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and
 - a base coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations, the base including
 - a body,
 - arms extending upwardly from the body and coupled to the fan housing on opposite sides of the airflow chamber, the arms defining a pivot axis for the fan housing, and
 - a stand having a first end pivotally coupled to the body and a second end opposite the first end, the stand being pivotable relative to the fan housing and the body between a collapsed position, in which the stand is flipped up toward the fan housing and the body such that the stand is positioned adjacent a side of the arms and the second end is positioned closer than the first end to the pivot axis, and a deployed position, in which the stand extends away from the fan housing and the body and is configured to contact the support surface;

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wherein the fan housing is movable relative to the body and the arms about the pivot axis.

2. The battery powered box-type fan assembly of claim 1, wherein the stand includes a pair of legs moveable with the stand.

3. The battery powered box-type fan assembly of claim 2, wherein the pair of legs are attached together by an elongated hinge member.

4. The battery powered box-type fan assembly of claim 1, wherein the stand is a first stand positioned on a first side of the base, and wherein the base further includes a second stand positioned on a second side of the base, the second stand being pivotable relative to the fan housing between a collapsed position, in which the second stand is flipped up toward the fan housing, and a deployed position, in which the second stand extends away from the fan housing and is configured to contact the support surface.

5. The battery powered box-type fan assembly of claim 4, wherein the first stand includes a first pair of legs, wherein the second stand includes a second pair of legs, wherein ends of the first pair of legs and the second pair of legs are rotatable downwardly and away from the fan housing when rotated from the collapsed position to the deployed position.

6. The battery powered box-type fan assembly of claim 5, wherein the first pair of legs and the second pair of legs are lockable into the deployed position.

7. The battery powered box-type fan assembly of claim 4, wherein the first stand and the second stand are rotatable relative to the fan housing independently from one another.

8. The battery powered box-type fan assembly of claim 1, wherein the base includes a latch coupled to the stand, the latch releasably securing a position of the stand, the latch being manually operable to allow movement of the stand relative to the fan housing.

9. The battery powered box-type fan assembly of claim 1, further comprising an AC power inlet, wherein the AC power inlet is electrically coupled to the motor, to the battery receptacle, or to both.

10. The battery powered box-type fan assembly of claim 9, wherein the AC power inlet is configured to provide charging power to the removable battery pack received in the battery receptacle.

11. The battery powered box-type fan assembly of claim 9, wherein the AC power inlet is positioned on the fan housing.

12. The battery powered box-type fan assembly of claim 1, wherein the fan housing further includes a first face adjacent an inlet side of the airflow chamber, and wherein the first face includes a retainer configured to receive and retain a replaceable filter on the first face.

13. The battery powered box-type fan assembly of claim 12, wherein the fan housing further includes a second face opposite the first face and adjacent an outlet side of the airflow chamber, and wherein the battery receptacle is accessible from the second face and located on a lower end of the fan housing.

14. The battery powered box-type fan assembly of claim 1, wherein the battery receptacle is configured to receive the removable battery pack in both the collapsed position and the deployed position.

15. A battery powered box-type fan assembly comprising:
a fan housing that at least partially defines an airflow chamber;
a plurality of fan blades positioned within the airflow chamber;
a motor operable to rotate the plurality of fan blades;

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a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and

a base coupled to the fan housing and configured to be placed on a support surface, the base including

a body elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations,

a stand movable relative to the body, and

a latch coupled to the body and the stand, the latch releasably securing a position of the stand relative to the body, the latch being manually operable to allow movement of the stand relative to the body;

wherein the latch includes a ratchet mechanism having a series of ramps and shoulders configured to contact a complementary series of ramps and shoulders formed on the stand to permit ratcheting rotation of the stand relative to the body in a first direction and inhibit rotation of the stand relative to the body in a second direction.

16. The battery powered box-type fan assembly of claim 15, wherein the stand is movable relative to the body between a collapsed position and a deployed position, wherein the first direction of the ratchet mechanism corresponds with rotation of the stand relative to the body from the collapsed position to the deployed position, and wherein the second direction of the ratchet mechanism corresponds with rotation of the stand relative to the body from the deployed position to the collapsed position.

17. A battery powered box-type fan assembly comprising:
a fan housing having a first face, a second face opposite the first face, and a perimeter made of a plurality of faces extending between the first and second faces; the first face, the second face, and the perimeter defining a volume therebetween, and the fan housing at least partially defines an airflow chamber;

a plurality of fan blades positioned within the airflow chamber;

a motor operable to rotate the plurality of fan blades;

a base pivotally coupled to the fan housing and configured to be placed on a support surface, the base including arms elevating the fan housing above the support surface, wherein the fan housing is pivotable relative to the arms about a pivot axis that is parallel to the support surface;

a battery receptacle positioned at a first corner formed by two adjacent faces of the perimeter of the fan housing below the pivot axis and electrically coupled to the motor;

an AC power inlet positioned at a second corner formed by two adjacent faces of the perimeter of the fan housing below the pivot axis and electrically coupled to the motor; and

a battery pack removably coupled to the battery receptacle, the battery pack being electrically coupled to the motor through the battery receptacle, the battery pack positioned within the volume of the fan housing;

wherein the battery pack rotates with the fan housing relative to the arms about the pivot axis when positioned within the volume of the fan housing; and

wherein the first face of the fan housing includes a retainer configured to selectively receive a replaceable filter and retain the replaceable filter on the fan housing, and wherein the battery pack is accessible on the second face of the fan housing.

18. The battery powered box-type fan assembly of claim 17, wherein the plurality of fan blades rotate about a rotation

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axis that is perpendicular to the pivot axis, and wherein the battery pack is further positioned beneath the rotation axis of the plurality of fan blades.

19. The battery powered box-type fan assembly of claim **17**, wherein the perimeter of the fan housing is polygonal. 5

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